Gap-Six Reconstruction, Complex Harmonic Determinants, and Stieltjes Boundary Collisions

Exact identities and reproducible proofs

Research continuation for the ProveIt project

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Abstract

We resolve three questions left open in the inspected ProveIt contributions and develop explicit identities from the resulting structure. First, every unequal even harmonic-tail pair with gap six has the optimal consecutive moment prefix. The proof makes the existing Gamma-basis asymptotic effective above an explicit threshold and certifies the entire finite complement by exact modular arithmetic. Second, formal harmonic degree equals generic pencil rank for every nonzero complex rational weight. A short rational difference theorem for symmetric matrix primitives removes the real-positivity hypothesis and allows varying pencil kernels. Third, we determine the full power–logarithm expansion, including the constant, for arbitrary products of argument derivatives of generalized Stieltjes constants whose positive shifts approach zero at unequal fixed rates.

The identity consequences include an all-order family of ordinary convergent rational polygamma-tail sums; a three-by-three Hahn formula for highest harmonic coefficients at arbitrary particle count; an exact classification of adjacent-pole residue exceptions; cutoff generators using only rational functions, logarithms, and one classical dilogarithm; and an explicit positive three-particle ensemble integral evaluated in ordinary zeta values. Proofs, finite exact certificates, and numerical diagnostics are distinguished throughout. The unrestricted all-pairs moment conjecture and the manuscript's short Gaussian S_6 and revised S_8 candidates remain open.

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1 Scope, source frontier, and principal results

This article continues the program in the two directories specified by the project owner: the polylogarithm manuscript and the incoming research contributions [1, 2]. All repository assertions refer to the pinned revision

`c4036d60715916ae72dd18983782fdb182840b86.`

The incoming index at that revision contains eleven archives. We extracted them, reconciled their overlapping research-status lists, and read the proof dependencies used below. This matters: an open question in an earlier contribution can already be solved in a later archive at the same revision.

In particular, the smaller-order-ten and smaller-order-twelve moment families had already been completed, as had the complete gap-two and gap-four families, several Stieltjes spectral faces, and arbitrary mixed even-tail products [4, 5, 3, 6]. Those results are inputs or context, and are not claimed anew. The useful remaining targets were the complete gap-six family, complex rational harmonic degree, and unequal-rate boundary collisions. The new theorems address these three targets directly.

1.1 What changes at this frontier

Question at the pinned revision	Result in this article	Precise scope
Optimal moment prefix for all even pairs with gap six	Theorem 2.1: the first q centered moments reconstruct all moments for $p = q - 6$, $q \geq 8$ even.	Full formal coefficient rank; an analytic infinite range plus an exact finite certificate.
Harmonic degree for complex rational weights	Theorem 3.3: degree equals generic pencil rank for every nonzero complex rational weight.	Rational dependence on cutoff; only finitely many admissible cutoff exceptions. No positivity hypothesis.
Explicit adjacent-pole determinants beyond small rank	Theorems 4.2 and 4.4: a three-by-three Hahn formula and a classical-polylogarithm generator for every particle count.	Specified two-pole family, with every endpoint residue included.
Unequal-rate Stieltjes boundary collisions	Theorems 5.1 and 5.2: all index and argument-derivative orders, with a finite contact formula for the constant.	Fixed positive rates, and locally uniform joint spectral subtraction.

Several identity consequences have a particularly short form. For example, with $H_M = \sum_{k=1}^{M} 1/k$, the positive ensemble in Section 4 satisfies

$$\sum_{0\leq n_1<n_2<n_3<M} \frac{\prod_{i<j}(n_j-n_i)^2}{\prod_i (n_i+1)(n_i+2)} = -\frac{M(M+1)(M+2)}{3}H_M^2 + \frac{4M^2(M+2)}{3}H_M + \frac{M^3(M^2-18M-31)}{12(M+1)}.$$

The boundary calculation produces the equally compact ordered identity

$$\mathrm{CT}_{\epsilon\downarrow 0}\int_0^1 \psi'(x+\lambda\epsilon)\psi(x+\mu\epsilon)\,dx = \zeta(2)\left(\frac{\mu}{\lambda}+\log\frac{\mu}{\lambda}\right), \qquad \lambda,\mu>0.$$

Here CT is explicitly the coefficient of $\epsilon^0(\log\epsilon)^0$, not the value of a divergent unsubtracted integral. The article supplies the complete subtractions as well as this constant.

1.2 The three mechanisms

The first mechanism turns a limiting determinant into an exact all-index result. The incoming Gamma basis has three exponential scales for gap six. The middle scale loses its leading imaginary phase, so a one-term Laplace estimate cannot prove the theorem. Retaining the next term produces a normalized reciprocal-square Vandermonde. A quantitative error comparison proves nonvanishing for every $b \geq 2048$, and exact recurrence arithmetic covers $4 \leq b \leq 2047$. Estimates are used only to finish an identity and reconstruction problem; no inequality is presented as the main research output.

The second mechanism is algebraic. If a symmetric rational matrix $G(z)$ has a rank-one difference

$$G(z) - G(z-1) = w(z)p(z)p(z)^{\mathsf{T}}$$

with $w \neq 0$ and independent rational components of p, then G cannot be generically singular. Otherwise its rank stabilizes at large integer cutoffs, equal ranks force its image to remain fixed, and the update vectors lie in a fixed proper subspace. That would make a nonzero constant combination of the components of p vanish identically. This elementary argument works over every field of characteristic zero, including complex fields with isotropic vectors.

The third mechanism is local at the singular endpoint. The Hurwitz recurrence splits each Stieltjes generating function into one explicit singular power and an analytic function at one. Only a critical Taylor coefficient of the analytic factor can contribute to the rate-dependent constant. The singular–singular term carries a beta or polylogarithmic kernel but contributes strictly negative powers of the collision parameter. The resulting constant is therefore much smaller than the full two-shift integral.

1.3 Conventions and logical status

We use $B_1 = -1/2$, rising factorials $(a)_k$, and falling factorials $a^{\underline{k}}$. Generalized harmonic numbers are $H_N^{(s)}(a) = \sum_{n=0}^{N}(n+a)^{-s}$, while the unshifted notation H_M runs from 1 to M. The polygamma convention is $\psi^{(r)} = d^{r+1}\log\Gamma/dz^{r+1}$, so $\psi^{(r)}(z) = (-1)^{r+1}r!\zeta(r+1,z)$ for $r \geq 1$. The generalized Stieltjes constants are fixed by

$$\zeta(1+u,z) = \frac{1}{u} + \sum_{m\geq 0}\frac{(-1)^m}{m!}\gamma_m(z)u^m.$$

Unless otherwise stated, argument derivatives are indicated by parenthesized superscripts; spectral derivatives are written explicitly.

Formal harmonic coordinates and formal zeta labels are independent symbols in their specified coefficient spaces. A rank theorem for that space is not a theorem of arithmetic independence for its evaluated constants. Similarly a cutoff generating function is first an identity on $|x| < 1$; a boundary value or an antiderivative at one is given its own convergence argument.

The article contains four kinds of evidence. Analytic and algebraic proofs establish the general statements. Exact modular nonvanishing is part of the gap-six proof on a finite, explicitly bounded interval. Other exact finite checks verify symbolic formulas and normalizations, without being substituted for general proofs. High-precision quadratures are diagnostics, with no claim that floating-point residuals are certified enclosures.

The methods of Christoffel transformations, Hahn polynomials, rational discrete residues, Euler sums, and local finite-part subtraction are established. They are credited to primary

references where used [10, 8, 9, 11, 12, 13]. The claimed advances are the precise theorems and specializations relative to the inspected ProveIt frontier. The literature review supports that distinction but is not an exhaustive historical-priority search. Independent internal proof audits are included; external peer review and proof-assistant formalization remain separate tasks.

2 The complete gap-six harmonic-tail reconstruction theorem

The fixed-gap Gamma basis of the incoming *Ordered Boundary Identities and Gamma Kernels* report [3] proves eventual reconstruction on every fixed-gap ray but does not specify a threshold. We make its argument effective when the gap is six and complete the remaining finite interval by exact modular arithmetic. This proves the entire previously unresolved gap-six family. The finite part is a computation-assisted proof: every determinant is certified by a nonzero residue in a finite field, with a short independently runnable certificate. No floating-point approximation enters that certificate.

2.1 Statement and the exact rational obstruction

Let $b \geq 4$ be an integer, $p = 2b - 6$, $q = 2b$, and

$$A_p(n) = \zeta(p) - H_n^{(p)} = \zeta(p, n+1), \qquad x_n = n + \tfrac{1}{2}.$$

Let $D_r = D_{p,q}(-2r)$, where

$$D_{p,q}(s) = \sum_{n \geq 0} A_p(n) A_q(n) x_n^{-s}$$

is initially defined in its convergence half-plane and subsequently continued meromorphically. The centered Hurwitz expansion shows that its possible poles occur at odd integers, so all displayed D_r are regular. The inherited moment blocks satisfy the inverse transforms of [3, Section 8]:

$$D_r = \sum_{j=0}^{r} \frac{\binom{2r}{2j}}{2j+1} B_{2r-2j}(1/2) C_j, \tag{2.1}$$

$$C_j = \sum_{r=0}^{j} \binom{2j+1}{2r} 2^{-2(j-r)} D_r. \tag{2.2}$$

For $j \geq b$, write

$$C_j = \sum_{k=0}^{b-1} a_{jk} Z_k, \qquad Z_k = \zeta(2k), \qquad Z_0 = -\tfrac{1}{2}. \tag{2.3}$$

With $d = 2j + 1$ and $e = d - p - q + 2k + 1$, these rational entries are zero when $e < 0$, and otherwise the exact late-coordinate formula of [3, Section 8] is

$$a_{jk} = B_e \left\{ \frac{\binom{d-q+1}{p-2k}}{d-q+1} + \frac{\binom{d-p+1}{q-2k}}{d-p+1} \right\}. \tag{2.4}$$

Every binomial summand whose lower index is not between 2 and its upper index is omitted. In particular a lower index of zero is omitted. Put

$$M_b = (a_{b+r,k})_{0 \leq r,k < b}.$$

All ranks below are ranks of rational coefficient matrices on the specified formal labels. They do not assert arithmetic independence of the actual zeta values.

Theorem 2.1 (Every gap-six pair has the optimal prefix)**.** *For every integer $b \geq 4$, the matrix M_b is nonsingular. Therefore*

$$\begin{aligned} \operatorname{span}_{\mathbb{Q}}\{D_0, D_1, \ldots\} &= \operatorname{span}_{\mathbb{Q}}\{D_0, \ldots, D_{2b-1}\} \\ &= \operatorname{span}_{\mathbb{Q}}\{C_0, \ldots, C_{b-1}, Z_0, \ldots, Z_{b-1}\}. \end{aligned} \tag{2.5}$$

The first $q = 2b$ moments have formal rank q, and their coefficient determinant in the displayed coordinate order is $\det M_b/(4b-1)!!$*. This number of consecutive moments is optimal for rank q on those q formal labels.*

Define, for $n \geq 1$,

$$\alpha_n = \frac{2|B_{2n}|}{(2n)!}, \qquad A_u = \begin{cases} 0, & 1 \leq u \leq 3, \\ (2u)^{\underline{6}}, & u \geq 4, \end{cases}$$

and, for $s \geq 4$,

$$B_s = (2s-1)^{\underline{6}}, \qquad d_s = 2(A_s + B_s) > 0.$$

Here the underline denotes a falling factorial. The B_s in this line are recurrence factors, distinguished from the Bernoulli numbers by the displayed definition. Define a rational array by

$$\begin{aligned} R_{s,j} &= \delta_{sj} && (1 \leq s, j \leq 3), \\ R_{s,j} &= \frac{1}{d_s} \sum_{u=1}^{s-1} \alpha_{s-u}(B_s + A_u) R_{u,j} && (s \geq 4,\ 1 \leq j \leq 3), \end{aligned} \tag{2.6}$$

and set

$$F_b = (R_{b+i,j})_{1 \leq i,j \leq 3}, \qquad f_b = \det F_b.$$

The inherited complementary-minor identity [3, Section 8, fixed-gap boundary reduction] is

$$\det M_b = \frac{(-1)^{b(b-1)/2}}{4^b b!} \left(\prod_{s=4}^{b+3} d_s \right) f_b. \tag{2.7}$$

Thus the theorem reduces exactly to $f_b \neq 0$ for every $b \geq 4$.

For clarity, the finite differential mechanism behind this reduction is the operator

$$\mathcal{T}R = (R^{(6)} - R^{(6)}(0)) \cot(z/2) + (R \cot(z/2))^{(6)}.$$

The coefficient of z^{2s-7} in $\mathcal{T}R$, for $R(z) = \sum_{u \geq 1} r_u z^{2u}$, is

$$d_s r_s - \sum_{u<s} \alpha_{s-u}(B_s + A_u) r_u.$$

The first three r_u are free and every later one is determined. Truncating these three null solutions at degree $2b$ and applying Jacobi's complementary-minor identity gives (2.7); the row and reversed-column factorial scales contribute $4^b b!$ and $(-1)^{b(b-1)/2}$, respectively. This is the same normalization as (2.4), including its omitted zero lower index.

2.2 The Gamma representation needed for an effective estimate

Put

$$W(x) = \left|\Gamma\left(\tfrac{1}{4} + \tfrac{ix}{2}\right)\right|^{12},$$
$$P_1(x) = \sinh(2\pi x), \qquad P_2(x) = \sinh(\pi x), \qquad P_3(x) = \tanh(\pi x),$$

and define

$$t_{n,j} = \frac{4}{(2n)!}\,\mathrm{Im}\int_0^\infty (x + i/2)^{2n} P_j(x) W(x)\,dx. \tag{2.8}$$

All these integrals converge absolutely. The inherited Gamma-basis result [3, Section 8, Gamma basis and positive normalization] says that the functions

$$G_j(z) = 4\sin(z/2)\int_0^\infty \sinh(xz) P_j(x) W(x)\,dx$$

satisfy $\mathcal{T}G_j = 0$, and their first three coefficient rows form a matrix $T = (t_{n,j})_{1\le n,j\le 3}$ with $\det T < 0$. Consequently

$$(t_{b+i,j})_{i,j=1}^3 = F_b T, \qquad f_b = \frac{\det(t_{b+i,j})_{i,j=1}^3}{\det T}. \tag{2.9}$$

These inputs have short verification mechanisms. The null equation follows from the Gamma recurrence after shifting two Fourier-integral contours by $\pm i$. The shifted common Gamma factor has its nearest poles at $\pm 3i/2$; only the tanh multiplier contributes the two simple poles at $\pm i/2$. Their residue terms provide exactly the subtracted $G_3^{(6)}(0)$, while the sinh multipliers contribute zero. For the initial determinant, $\mathrm{Im}(x+i/2)^{2n}$ has leading term nx^{2n-1}. Continuous Cauchy–Binet therefore reduces its sign to the product of the two alternants

$$\det(x_r^{2n-1})_{n,r=1}^3, \qquad \det(P_j(x_r))_{r,j=1}^3.$$

After extracting $\prod_r x_r \tanh(\pi x_r)$, the first is a Vandermonde in x_r^2 and the second is minus twice a Vandermonde in $\cosh(\pi x_r)$. Both variable maps are strictly increasing on $(0,\infty)$; integration against $\prod_r W(x_r)$ is therefore strictly negative. This proves $\det T < 0$ without a numerical normalization constant.

2.3 An explicit density bound

Let

$$\lambda_j = j\pi, \qquad A = 8(2\pi)^6, \qquad a_1 = a_2 = A/2, \quad a_3 = A. \tag{2.10}$$

Lemma 2.2 (Uniform Gamma density estimate)**.** *For every $x \ge 2$ and $1 \le j \le 3$,*

$$P_j(x)W(x) = a_j x^{-3} e^{-\lambda_j x}(1 + \delta_j(x)), \qquad |\delta_j(x)| \le \frac{6}{x^2}. \tag{2.11}$$

Moreover $a_j > 1$, and on $0 \le x \le 2$,

$$0 \le P_j(x)W(x)/a_j < 2^{54}. \tag{2.12}$$

Proof. We use the classical complex Stirling remainder in DLMF §5.11(ii). For $z = 1/4 + ix/2$, $|\arg z| < \pi/2$, so retaining $1/(12z)$ in Stirling's logarithmic expansion gives

$$\log\Gamma(z) = (z - \tfrac{1}{2})\log z - z + \tfrac{1}{2}\log(2\pi) + \frac{1}{12z} + \rho(z), \qquad |\rho(z)| \le \frac{1}{90|z|^3}.$$

Indeed the first omitted term has magnitude $1/(360|z|^3)$, and the secant factor is at most 4. Set $t = 1/(2x)$ and

$$L(x) = \log \frac{W(x)}{Ax^{-3}e^{-3\pi x}}.$$

Taking twelve times the real part and simplifying the elementary logarithm yields

$$L(x) = -\frac{3}{2}\log(1+t^2) + 3\left(\frac{\arctan t}{t} - 1\right) + \frac{4t^2}{1+t^2} + 12\,\mathrm{Re}\,\rho(z).$$

The inequalities $\log(1+t^2) \le t^2$ and $0 \le 1 - \arctan(t)/t \le t^2/3$ give

$$|L(x)| \le \frac{13}{8x^2} + \frac{16}{15x^3} \le \frac{259}{120x^2} < \frac{9}{4x^2} \qquad (x \ge 2).$$

Since $e^{9/16} < 2$,

$$\left|\frac{W(x)}{Ax^{-3}e^{-3\pi x}} - 1\right| \le \frac{9}{2x^2}.$$

The elementary multipliers differ from their leading exponentials by at most $2e^{-2\pi x}$. More precisely their normalized factors are $1 - e^{-4\pi x}$, $1 - e^{-2\pi x}$, and $(1 - e^{-2\pi x})/(1 + e^{-2\pi x})$, respectively, and each lies between zero and one. Since $\pi > 3$ and $e^{6x} \ge (6x)^2/2$,

$$2e^{-2\pi x} \le \frac{1}{9x^2}.$$

Combining the two errors proves (2.11), with room to spare.

Euler's Gamma integral gives $|\Gamma(1/4 + ix/2)| \le \Gamma(1/4) < 5$, because the contributions from $(0,1)$ and $(1,\infty)$ are less than 4 and 1. For $0 \le x \le 2$ all three multipliers are less than e^{16}. Using $e < 3$, $5^{12} < 2^{28}$, $3^{16} < 2^{26}$, and $a_j > 1$ proves (2.12). □

2.4 A completely explicit shifted Laplace estimate

For an even integer $N \ge 4096$ and $\lambda \in \{\pi, 2\pi, 3\pi\}$, write

$$S_N(\lambda) = \frac{\Gamma(N-2)}{\lambda^{N-2}}, \quad S_N^{(1)}(\lambda) = \frac{\Gamma(N-3)}{\lambda^{N-3}}, \quad S_N^{(2)}(\lambda) = \frac{\Gamma(N-4)}{\lambda^{N-4}}.$$

Lemma 2.3 (Effective two-term phase estimate)**.** *For $1 \le j \le 3$ and every even $N \ge 4096$,*

$$\left|\frac{1}{a_j}\int_0^\infty (x + i/2)^N P_j(x) W(x)\,dx - e^{i\lambda_j/2}\left(S_N(\lambda_j) + \frac{3i}{2}S_N^{(1)}(\lambda_j)\right)\right| \le 16 S_N^{(2)}(\lambda_j). \tag{2.13}$$

Proof. The contribution of $[0,2]$ is at most $2^{55}3^N$ by (2.12). On $[2,\infty)$ substitute the model $x^{-3}e^{-\lambda x}$ and use (2.11). The density error is at most

$$6\int_2^\infty |x + i/2|^N x^{-5} e^{-\lambda x}\,dx.$$

On $x \ge \sqrt{N}$,

$$|x + i/2|^N \le x^N \exp\left(\frac{N}{8x^2}\right) < 2x^N,$$

so this part is at most $12 S_N^{(2)}(\lambda)$. On $2 \le x \le \sqrt{N}$, its bound is

$$\sqrt{N}(N+1)^{N/2}.$$

Here we used $6x^{-5} \le 6/32 < 1$ and the length at most $\sqrt{N}$.

For the model integral set $y = x + i/2$ and shift the ray to the positive real axis beginning at 2. The integrand

$$e^{i\lambda/2} y^N (y - i/2)^{-3} e^{-\lambda y}$$

has no pole in this strip. The short vertical segment at 2 is bounded by $3^N/16$, and the segment at infinity vanishes. For real $y \geq 2$, the binomial series gives

$$\left| (1 - i/(2y))^{-3} - 1 - \frac{3i}{2y} \right| \leq \frac{5}{2y^2}.$$

To check the constant, write $u = 1/(2y) \leq 1/4$; then $\sum_{k\geq 2} \binom{k+2}{2} u^k \leq 10u^2$ since its quotient by u^2 is increasing and is less than 10 at $u = 1/4$. Thus this model error is at most $(5/2)S_N^{(2)}(\lambda)$. Extending the first two retained integrals from $[2, \infty)$ to $(0, \infty)$ contributes less than 2^N.

It remains to absorb the compact and lower-range contributions. Their sum is bounded by

$$R_N = 2^{56} 3^N + \sqrt{N}(N+1)^{N/2}.$$

Since $\lambda < 10$ and $(N-5)! \geq ((N-5)/3)^{N-5}$,

$$S_N^{(2)}(\lambda) \geq \frac{(N-5)^{N-5}}{3^{N-5} 10^{N-4}}.$$

For every $N \geq 4096$,

$$\frac{30\sqrt{N+1}}{N-5} \leq \frac{1}{2}, \qquad \frac{90}{N-5} \leq \frac{1}{32}, \qquad N^6 \leq 2^{N-2}.$$

The first follows by squaring and expanding a quadratic; the second is immediate; the third holds at $N = 4096$ and persists by induction because $(1 + 1/N)^6 < 2$. Therefore

$$\frac{\sqrt{N}(N+1)^{N/2}}{S_N^{(2)}} \leq N^6 2^{-N} \leq \frac{1}{4}, \qquad \frac{2^{56} 3^N}{S_N^{(2)}} \leq 2^{56} N^5 2^{-5N} < \frac{1}{4}.$$

Consequently $R_N < S_N^{(2)}$. Adding the bounds gives $12 + 5/2 + 1 < 16$, proving (2.13). □

The retained second term is essential. At $j = 2$ the leading phase $e^{i\lambda_j/2} = -1$ has zero imaginary part, so discarding that second term would give no nonzero leading coefficient for the middle column.

2.5 A determinant comparison valid for every large index

Taking imaginary parts of (2.13) gives, with $N = 2n \geq 4096$,

$$t_{n,1} = \frac{4a_1}{N!} S_N(\lambda_1)(1 + \epsilon_{n,1}),$$

$$t_{n,3} = -\frac{4a_3}{N!} S_N(\lambda_3)(1 + \epsilon_{n,3}), \qquad |\epsilon_{n,j}| \leq \frac{1600}{(N-4)^2} \quad (j = 1, 3), \tag{2.14}$$

$$t_{n,2} = -\frac{6a_2}{N!} S_N^{(1)}(\lambda_2)(1 + \epsilon_{n,2}), \qquad |\epsilon_{n,2}| \leq \frac{320}{3(N-4)}. \tag{2.15}$$

The estimates use only $\lambda_j < 10$ and the exact Gamma ratios.

Let $b \geq 2048$ and $n = b + i$ for $1 \leq i \leq 3$. After extracting positive row and column factors and the column signs $(+, -, -)$, the coefficient determinant is the determinant of an array $E = (E_{ij})$ satisfying

$$E_{ij} = j^{-2i}(1 + \eta_{ij}), \qquad |\eta_{ij}| < \frac{1}{25}. \tag{2.16}$$

For precision, extract the positive row factors $4\Gamma(2b+2i-2)\pi^{-2i}/\Gamma(2b+2i+1)$ and positive column factors $a_j\lambda_j^{-2b+2}$ for $j=1,3$. For $j=2$, multiply this column factor additionally by $3\lambda_2/(2(2b-1))$. The middle-column row correction is

$$\rho_i=\frac{2b-1}{2b+2i-3},\qquad 0<\rho_i\le 1,\qquad 1-\rho_i\le\frac{4}{2b-1}.$$

Thus its relative error is bounded by

$$\frac{4}{2b-1}+\frac{320}{3(2b-2)}<\frac{1}{25}\qquad (b\ge 2048),$$

while the outer-column bound follows at once from (2.14). The product of the three extracted signs is positive.

For the exact limiting array $V_{ij}=j^{-2i}$ one computes

$$\det V=-\frac{5}{1944},\qquad \operatorname{perm} V=\frac{289}{23328}. \tag{2.17}$$

The permanent here is the sum of the six unsigned permutation products. By expanding both determinants,

$$|\det E-\det V|\le \operatorname{perm} V\left\{(26/25)^3-1\right\}=\frac{289}{23328}\frac{1951}{15625}<\frac{5}{1944}.$$

The exact positive margin is $373661/364500000$. Therefore $\det E<0$, and hence

$$f_b>0\qquad (b\ge 2048) \tag{2.18}$$

by (2.9) and $\det T<0$. This is an effective all-index conclusion, rather than a numerical comparison at finitely many large indices.

2.6 Exact finite completion

The remaining range is $4\le b\le 2047$. The following two elementary recurrences give a finite certificate without computing huge rational Bernoulli numbers. From

$$\frac{x}{2}\cot\frac{x}{2}=1-\frac{1}{2}\sum_{n\ge1}\alpha_n x^{2n}$$

and the differential equation of the cotangent, one obtains

$$\alpha_1=\frac{1}{6},\qquad \alpha_n=\frac{1}{2(2n+1)}\sum_{k=1}^{n-1}\alpha_k\alpha_{n-k}\quad (n\ge2). \tag{2.19}$$

Combine this recurrence with (2.6) modulo

$$\ell=2147483647=2^{31}-1.$$

The supplied script proves primality by trial division through $\lfloor\sqrt{\ell}\rfloor$. Every denominator is invertible in this field: for the needed indices $n\le2049$ the factors $2(2n+1)$ are smaller than ℓ, and

$$d_s=2\frac{(2s)!}{(2s-6)!}\frac{2s-3}{s}$$

for $4\le s\le2050$ has no factor divisible by ℓ. Induction therefore identifies every computed residue with the reduction of its exact rational coefficient.

Proposition 2.4 (Finite rational nonvanishing certificate)**.** *For each integer $4 \le b \le 2047$, the determinant f_b has a nonzero reduction modulo* 2147483647*. Hence $f_b \neq 0$ over $\mathbb{Q}$ throughout this finite interval.*

Exact computational proof. Run `verify_gap_six.py`. It constructs all needed α_n by (2.19), constructs the three fundamental columns by (2.6), and expands each three-by-three determinant by its six signed products. All 2,044 residues are nonzero; the complete indexed list is in `gap_six_certificate.json`. Its canonical compact JSON residue list has SHA-256

`90ec1b66f919cd00404dd92f7d1f9483ceb9`
`d53bfa5893735db62c797113fea6`

The hash identifies the record, while the recurrences and the nonzero residues constitute the mathematical certificate.

As independent normalization checks, all 2,049 cotangent coefficients agree with formal division of cosine by the normalized sine series, and the bridge (2.7) agrees with direct exact determinants of the original Bernoulli matrices for $4 \le b \le 20$. The latter comparisons check the omitted-index convention and every row/column factor against (2.4). They are additional checks; the finite-field recurrence itself proves the entire finite interval. □

Combining (2.18), Proposition 2.4, and (2.7) proves Theorem 2.1. The general unrestricted unequal-even-order conjecture remains open beyond the families now covered. In particular, proving all gaps of six does not by itself prove the gap-eight family.

2.7 An all-order family of rational polygamma-tail evaluations

The rank theorem turns the source block calculus into an exact ordinary convergent identity for every gap-six pair. Let $(\lambda_b, \dots, \lambda_{2b-1})$ be the first row of M_b^{-1}, and put

$$K_b(y) = \sum_{j=b}^{2b-1} \lambda_j y^{2j+1},$$

$$P_b(x) = K_b(x + \tfrac{1}{2}) - K_b(x - \tfrac{1}{2}) = \sum_{r=0}^{q-1} w_r x^{2r}. \tag{2.20}$$

All coefficients are explicitly specified rational numbers; the inverse exists for every $b \ge 4$ by Theorem 2.1. For $h = 0, 1, 2, 3$ define

$$c_{v,0} = \frac{1}{v-1}, \qquad c_{v,h} = \frac{B_{2h}(1/2)}{(2h)!}(v)_{2h-1} \quad (h \ge 1), \qquad u_h = \sum_{e=0}^{h} c_{p,e} c_{q,h-e}.$$

Here $(v)_r$ denotes the rising factorial. Define the even sextic subtraction polynomial

$$S_b(x) = \sum_{h=0}^{3} \left(\sum_{e=0}^{h} u_e \, w_{q-1-h+e} \right) x^{6-2h}. \tag{2.21}$$

Corollary 2.5 (Rational evaluations for every even gap of six)**.** *For $b \ge 4$, $p = 2b-6$, $q = 2b$, and $x_n = n + 1/2$,*

$$\boxed{\sum_{n=0}^{\infty} \left\{ \frac{P_b(x_n)}{(p-1)!(q-1)!} \psi^{(p-1)}(n+1)\psi^{(q-1)}(n+1) - S_b(x_n) \right\} = -\frac{1}{2}.} \tag{2.22}$$

This sum converges absolutely. Its polynomial P_b has exact degree $2q-2$, with leading coefficient

$$w_{q-1} = \frac{4b(4b-1) f_{b-1}}{d_{b+3} f_b} \neq 0, \tag{2.23}$$

where the auxiliary initial determinant is $f_3 = -1/638788869734400000$.

Proof. The first inverse row gives $\sum_{j=b}^{2b-1} \lambda_j C_j = Z_0 = -1/2$. The coefficients of the centered difference in (2.20) are exactly

$$w_r = \sum_{j=\max\{b,r\}}^{2b-1} \lambda_j \binom{2j+1}{2r} 2^{-2(j-r)},$$

so (2.2) gives $\sum_r w_r D_r = -1/2$. The centered tail expansion gives

$$A_p(n)A_q(n) = \sum_{h=0}^{3} u_h x_n^{2-p-q-2h} + O(x_n^{-p-q-6}).$$

Since $\deg P_b \le 2q-2$ and $q-p=6$, its nonnegative powers are exactly (2.21); the remaining summand is $O(x_n^{-2})$. The continued sums of $1, x_n^2, x_n^4, x_n^6$ are zero because $\zeta(-2h, 1/2) = 0$. Thus the ordinary absolutely convergent sum equals the displayed moment combination. Finally $A_p(n) = \psi^{(p-1)}(n+1)/(p-1)!$ for even p.

For (2.23), suppose first that $b \ge 5$. The cofactor defining $(M_b^{-1})_{0,b-1}$ deletes the final row and the first column of M_b. Reindexing $j' = j-1$, $k' = k-1$, $p' = p-2$, and $q' = q-2$ in (2.4) shows that this minor is exactly M_{b-1}: both binomial pairs, the Bernoulli index, and the omitted-index convention are unchanged. Hence

$$\lambda_{2b-1} = (-1)^{b-1} \frac{\det M_{b-1}}{\det M_b} = \frac{4b f_{b-1}}{d_{b+3} f_b}$$

by (2.7). The centered difference multiplies its highest coefficient by $4b-1$, proving the formula and nonvanishing for $b \ge 5$. At $b = 4$ a direct exact calculation gives $\lambda_7 = 84/83$, $d_7 = 6795360$, $f_4 = -83/22789201752651350016000000$, and the stated value of f_3; these verify the same formula without invoking a zero-order tail. This completes the proof. □

Further research. The same Gamma basis supplies m distinct exponential scales for any fixed half-gap m. Making all of its phase estimates effective would give a numerical finite-completion threshold for each fixed gap. The next concrete target is $m = 4$, where the two even modes require their first nonzero phase corrections and the determinant comparison involves a four-by-four reciprocal-square Vandermonde. A different question is whether a direct recurrence invariant can prove gap-six nonvanishing without the finite completion; the initial sign transition shows why a fixed-sign assertion from the first admissible index is false.

3 Complex rational weights: a complete degree–rank theorem

The recent *Bilinear Tails and Harmonic Determinants* contribution proves that the formal harmonic degree of a moment determinant equals its generic harmonic-pencil rank for every nonzero real rational weight. It explicitly leaves the corresponding assertion for complex rational weights open [4, Section 5, Corollary `deg:rational`; Section 10, "Complex weights and sharp cutoff exceptions"]. We prove the complex assertion. The new argument uses rationality in the cutoff and the rigidity of symmetric rank-one updates. It uses ordinary transposition throughout; no Hermitian positivity is assumed. The stronger numerical bound on the number of exceptions for eventually one-sign real weights remains a separate advantage of the source theorem.

3.1 A rational difference principle

All fields in the next statements have characteristic zero. The indeterminate is z, and linear independence of functions means independence over the constant field, unless another field is displayed.

Lemma 3.1 (Equal ranks force equal images)**.** *Let A be a symmetric $q \times q$ matrix over a field F, let $c \in F \setminus \{0\}$, and let $u \in F^q$. If $u \notin \operatorname{im} A$, then*

$$\operatorname{rank}(A + cuu^{\mathsf{T}}) = \operatorname{rank} A + 1.$$

Consequently, if A and $A + cuu^{\mathsf{T}}$ have equal rank, then $u \in \operatorname{im} A$ and $\operatorname{im}(A + cuu^{\mathsf{T}}) = \operatorname{im} A$.

Proof. Put $d = \operatorname{rank} A$. Symmetric elimination gives an invertible matrix S for which

$$S^{\mathsf{T}} A S = \begin{pmatrix} A_0 & 0 \\ 0 & 0 \end{pmatrix}, \qquad \det A_0 \neq 0.$$

Write $S^{\mathsf{T}} u = (v, w)^{\mathsf{T}}$. If u is outside the image, some coordinate w_j is nonzero. The principal minor on the d coordinates of A_0 and this one further coordinate is

$$\det \begin{pmatrix} A_0 + cvv^{\mathsf{T}} & cvw_j \\ cw_j v^{\mathsf{T}} & cw_j^2 \end{pmatrix} = cw_j^2 \det A_0 \neq 0. \tag{3.1}$$

For instance, subtract (v_i/w_j) times the last row from row i to verify the equality directly. A rank-one update can increase rank by at most one, proving the first claim. Equal ranks therefore force $u \in \operatorname{im} A$. In that case the new image is contained in the old one and has the same dimension, proving equality. The proof remains valid when $w^{\mathsf{T}} w = 0$; nonzero complex isotropic vectors cause no exception to the argument. □

Theorem 3.2 (Nonsingularity of rational symmetric primitives)**.** *Let $G(z) \in \operatorname{Sym}_q(F(z))$, let $w(z) \in F(z)$ be nonzero, and let $p(z) \in F(z)^q$ have linearly independent components over F. If*

$$G(z) - G(z-1) = w(z) p(z) p(z)^{\mathsf{T}}, \tag{3.2}$$

then $\det G(z)$ is a nonzero rational function. This conclusion holds for every symmetric rational solution, including after adding an arbitrary constant symmetric matrix.

Proof. Suppose the generic rank were $d < q$. A nonzero d-minor of G has finitely many zeros and poles, and every $(d+1)$-minor vanishes identically. Hence there is an integer N_0 such that, at every integer $N \geq N_0$, the matrices $G(N)$ and $G(N-1)$ are defined and have rank d, while $w(N) \neq 0$ and $p(N)$ is defined. Characteristic zero ensures that these are infinitely many distinct evaluation points. Lemma 3.1 applied to (3.2) shows that the images of all these matrices are one fixed d-dimensional subspace $W \subset F^q$, and that $p(N) \in W$ at every sufficiently large integer.

Choose a nonzero linear functional $\lambda \in (F^q)^*$ vanishing on W. Then the rational function $\lambda p(z)$ vanishes at infinitely many distinct points, so it is identically zero. This contradicts independence of the components of p. Thus $d = q$. Adding a constant symmetric matrix preserves (3.2), so the same proof applies to every such addition. □

The result also applies with any fixed nonzero translation h in place of 1, by changing variables. Rationality is used to make the ranks constant outside a finite set and to turn infinitely many zeros into a function identity. The statement is consequently stronger than a positivity argument, but it does not make a claim about arbitrary nonrational sequences of bilinear Gram matrices.

3.2 The complex harmonic-degree theorem

Let $C_1, \dots, C_t \in \mathrm{Sym}_r(F)$ and put

$$C(X) = \sum_{j=1}^{t} X_j C_j, \qquad \rho = \mathrm{rank}_{F(X)}\, C(X), \qquad q = r - \rho.$$

Suppose $R \in F(z)$ is nonzero and that the finite moment matrices admit a decomposition

$$M_N = \sum_{n=0}^{N} R(n) v(n) v(n)^{\mathsf{T}} = D(N) + \sum_{j=1}^{t} h_j(N) C_j, \qquad v(n) = (1, n, \dots, n^{r-1})^{\mathsf{T}}, \tag{3.3}$$

where $D(z) \in \mathrm{Sym}_r(F(z))$. It suffices that the decomposition and all its terms be defined for every sufficiently large integer N. The values $h_j(N)$ may lie in an extension of F. The variables X_j are formal and independent; no arithmetic independence of the actual harmonic values is asserted.

Theorem 3.3 (Complex rational degree–rank equality)**.** *Under these hypotheses,*

$$\boxed{\deg_X \det(D(z) + C(X)) = \rho \quad \textit{over } F(z).} \tag{3.4}$$

Only finitely many admissible integer cutoffs can have smaller degree. In particular the conclusion holds for every nonzero complex rational weight with no poles at nonnegative integers and its usual finite harmonic decomposition.

Proof. The determinant has degree at most ρ, since any term choosing more than ρ columns from $C(X)$ is zero. Since F is infinite, choose $a \in F^t$ for which $A = C(a)$ has rank ρ. Let K be an $r \times q$ constant matrix whose columns form its kernel. Choose a complement U so that $S = [U\ K]$ is invertible and

$$S^{\mathsf{T}} A S = \begin{pmatrix} A_0 & 0 \\ 0 & 0 \end{pmatrix}, \qquad \det A_0 \neq 0.$$

Every matrix $A + tC_j$ has rank at most ρ. Over $F(t)$ its upper-left block is invertible near $t = 0$, so its Schur complement vanishes identically. The coefficient of t in that complement is $K^{\mathsf{T}} C_j K$. Therefore

$$K^{\mathsf{T}} C_j K = 0 \qquad (1 \le j \le t). \tag{3.5}$$

This algebraic argument is the complex form of the source's compression lemma.

Put $G(z) = K^{\mathsf{T}} D(z) K$ and $p(z) = K^{\mathsf{T}} v(z)$. The components of p are independent polynomials, because K has column rank q. Equations (3.3) and (3.5) show, at all sufficiently large integers, that

$$G(N) - G(N-1) = R(N) p(N) p(N)^{\mathsf{T}}.$$

Both sides are rational functions in N, so this is the identity (3.2). Theorem 3.2 gives $\det G(z) \neq 0$. Finally,

$$[t^{\rho}] \det(D(z) + tA) = \frac{\det A_0}{(\det S)^2} \det G(z) \neq 0, \tag{3.6}$$

obtained by selecting tA_0 in its ρ rows and columns after congruence by S. This proves (3.4). The same coefficient is nonzero at all but finitely many admissible integer cutoffs, proving the final assertion. When $q = 0$, its determinant is one and the conclusion is immediate. □

No common kernel of the entire pencil is assumed. The kernel of one maximal-rank member is only totally isotropic for every C_j, which is exactly the compression property needed in the proof. The theorem answers the complex-weight question affirmatively rather than restricting it to pencils with a common kernel.

3.3 Finite certificates for every exceptional cutoff

The proof gives a cheap sufficient certificate using just one maximal-rank direction. Choose a nonzero polynomial $d(z)$ clearing the denominators of $G(z)$ and set $H(z) = d(z)G(z)$. Then

$$0 \neq E(z) := \det H(z) \in F[z]. \tag{3.7}$$

Every degree-deficient cutoff at which $d(N) \neq 0$ is a root of E. If $e_{ij} = \deg H_{ij}$, with degree $-\infty$ for zero entries, then

$$\deg E \leq \max_{\pi \in S_q} \sum_{i=1}^{q} e_{i,\pi(i)} \leq q \max_{i,j} e_{ij}. \tag{3.8}$$

Thus the right side bounds the number of distinct such exceptions. This is a bound computable from rational entries and does not depend on ordering the coefficient field. Apparent poles introduced by denominator clearing must be removed or checked in the original decomposition.

For an exact rather than sufficient certificate, write the highest homogeneous component after clearing a common denominator in z as

$$[\det(D(z) + C(X))]_\rho = \frac{1}{b(z)} \sum_{|\nu|=\rho} e_\nu(z) X^\nu.$$

Theorem 3.3 guarantees that not all e_ν vanish. For an admissible N with $b(N) \neq 0$, degree is deficient exactly when $g(N) = 0$, where

$$g(z) = \gcd_{|\nu|=\rho} e_\nu(z). \tag{3.9}$$

This is the complex-field extension of the source's coefficient-gcd test; it is not a new polynomial-gcd algorithm. The cheaper polynomial E supplies a finite candidate set, and the full highest component filters it exactly.

3.4 A residue interpretation of the rational compression

For clarity, the usual harmonic decomposition is available over $\mathbb{C}$ without a sign assumption. Partial fractions of $z^m R(z)$ consist of a polynomial and finitely many terms $c/(z-\alpha)^j$. Poles differing by integers belong to one shift orbit. Summing those terms from 0 to N produces one generalized harmonic coordinate per orbit and pole order, together with rational endpoint corrections. This is the form (3.3).

The discrete residue of a rational function at an orbit and pole order is the sum of the corresponding partial-fraction coefficients over that orbit. The classical rational-summability criterion states that a rational function has a rational antidifference if and only if all these discrete residues vanish [9, Proposition 2.5]. For a polynomial subspace V with basis $p_1, \ldots, p_q$, the following statements are equivalent:

1. every $R(z)p_i(z)p_j(z)$ has a rational antidifference;
2. every discrete-residue bilinear form $(f, g) \mapsto \operatorname{dres}_{\mathcal{O},k}(Rfg)$ vanishes on $V \times V$;
3. if K is the coefficient matrix of this basis, then $K^{\mathsf{T}} C_j K = 0$ for every harmonic coefficient matrix.

The equivalence is the classical residue criterion applied entry by entry. Theorem 3.2 adds the following consequence: whenever these conditions hold and the p_i are independent, every symmetric rational matrix primitive of the products Rp_ip_j is nonsingular over $\mathbb{C}(z)$. Thus a maximal-rank pencil kernel produces a rationally summable polynomial space whose accumulated bilinear form cannot be identically degenerate.

3.5 Exact nonreal examples

Let

$$R_\lambda(z) = \frac{1}{z+1} + \frac{\lambda}{z+2}, \qquad \lambda \neq 0, \qquad X = \sum_{n=0}^{N} \frac{1}{n+1}.$$

For an $r \times r$ moment matrix, $r \geq 2$, the only harmonic coefficient matrix is

$$C = u_1 u_1^{\mathsf{T}} + \lambda u_2 u_2^{\mathsf{T}}, \qquad u_a = (1, -a, \ldots, (-a)^{r-1})^{\mathsf{T}},$$

and has rank two. Thus the harmonic degree is exactly two over $\mathbb{C}(N)$, for every nonzero complex λ.

At $r = 3$, the kernel polynomial is $(z+1)(z+2)$ and

$$\boxed{[X^2]\det(D(N) + XC) = \frac{\lambda}{12}(N+1)(N+2)(N+3)\{3(\lambda+1)N + 4\lambda + 8\}.} \tag{3.10}$$

At $r = 4$, use the two kernel polynomials $(z+1)(z+2)$ and $z(z+1)(z+2)$. Then

$$\begin{aligned}[X^2]\det(D(N) + XC) = \frac{\lambda}{3600} N(N+1)^2(N+2)^2(N+3)^2(N+4) \\ \cdot \{6(\lambda+1)^2N^2 + (14\lambda^2 + 48\lambda + 34)N + 10\lambda^2 + 45\lambda + 50\}.\end{aligned} \tag{3.11}$$

Both identities follow from (3.6) and finite sums of polynomials. In the four-dimensional case, the two-by-two compressed determinant is the expression after the initial factor λ. For $\lambda = i$, its final quadratic is $20N + 40 + i(12N^2 + 48N + 45)$, so it cannot vanish at a nonnegative integer. Consequently the four-dimensional harmonic degree is exactly two at every $N \geq 1$, even though the actual evaluated moment determinant vanishes at $N = 1, 2$ because too few nodes are present. Formal degree and numerical determinant value remain different notions.

A nonreal rational weight can nevertheless have a genuine later degree exception. Take $r = 4$ and

$$R(z) = \frac{39}{z+1} + \frac{78i}{z+2} - \frac{61 + 95i}{z+3}. \tag{3.12}$$

Its one harmonic matrix has rank three. The kernel polynomial is $p(z) = (z+1)(z+2)(z+3)$, and exact Faulhaber summation gives

$$\begin{aligned}\sum_{n=0}^{N} R(n)p(n)^2 = -\frac{1}{6}(N-3)(N+1)(N+2)(N+3)(N+4) \\ \cdot \{(22 + 17i)N + 56 + 22i\}.\end{aligned} \tag{3.13}$$

The top-left three-square minor of the harmonic matrix is $1155960 - 742248i \neq 0$. Hence the only degree-deficient nonnegative cutoff is $N = 3$. At that cutoff the formal determinant is exactly

$$\begin{aligned}P_3(X) = (-22046418576 + 2049048144i)X^2 \\ + \left(\frac{455872142232}{5} - 7792299840i\right) X \\ - \frac{2356478577592}{25} + \frac{183512850456}{25} i.\end{aligned} \tag{3.14}$$

Thus its degree drops from three to two, while substitution of the actual $X = 25/12$ gives

$$P_3(25/12) = \frac{55033 - 146419i}{25} \neq 0.$$

Independently, this value is the four-node Vandermonde square times $R(0)R(1)R(2)R(3)$. The exception is therefore a formal-degree phenomenon at a nonzero evaluated determinant, and not a lack of nodes or a vanishing weight.

3.6 Integration and further questions

The source's real-weight results are valid and need no retraction. Its open complex-weight question can be marked solved under the same rational-decomposition hypotheses, and Corollary `deg:rational` can be broadened to complex rational weights. The eventual-sign bound $r(r-\rho)$ should retain its original real hypotheses. The present denominator-degree certificate is a different bound and is not asserted to equal that real bound.

Two precise next questions remain. First, determine sharp bounds for the exceptional-cutoff polynomial using only the pole orders and integer dispersions of R, without constructing the whole compressed primitive. Second, classify polynomial subspaces totally isotropic for all discrete-residue forms of Rfg; the primitive theorem proves nonsingularity once such a subspace is given, but does not classify the subspaces or the rank-two pencils producing them.

4 Adjacent poles, Hahn determinants, and classical polylogarithms

The complex degree theorem explains why a rational weight may have surprisingly small harmonic complexity even in a large determinant. We now make that statement explicit for two adjacent simple poles. The general Christoffel determinant transformation used below is classical; its particularly compact form and its harmonic consequences are the specialization developed here. We use the formulation in [10] and give the short determinant proof needed for our normalization. The finite orthogonal polynomials are the uniform Hahn, or discrete Chebyshev, polynomials [8, Sections 18.19–18.20].

4.1 A three-by-three formula for every particle count

For $r \geq 2$ and an admissible shift a, set

$$R_\eta(n;a) = \frac{1}{n+a} + \frac{\eta}{n+a+1},$$
$$Z_{r,N}(a,\eta) = \det_{0\leq i,j<r}\left(\sum_{n=0}^{N} n^{i+j} R_\eta(n;a)\right). \tag{4.1}$$

Admissible means that no displayed denominator vanishes. For $N \geq r-1$, finite Cauchy–Binet gives the equivalent ensemble

$$Z_{r,N}(a,\eta) = \sum_{0\leq n_1<\cdots<n_r\leq N} \prod_{i<j}(n_j-n_i)^2 \prod_{\ell=1}^{r} R_\eta(n_\ell;a). \tag{4.2}$$

An empty sum is zero. Resolve the shifted harmonic coordinate by

$$X = H_N^{(1)}(a) := \sum_{n=0}^{N} \frac{1}{n+a}, \qquad \sum_{n=0}^{N} \frac{1}{n+a+1} = X + \frac{1}{N+a+1} - \frac{1}{a}.$$

The determinant becomes a formal polynomial $\mathcal{P}_{r,N}(X;a,\eta)$ with rational coefficients in the cutoff and parameters. For $\eta \neq 0$ its harmonic pencil has rank two, so its generic degree in X is exactly two. Write $T_{r,N}(a,\eta) = [X^2]\mathcal{P}_{r,N}$.

Proposition 4.1 (A polynomially weighted compression)**.** *Put $q = r-2$ and $L_a(n) = (n+a)(n+a+1)$. Then*

$$\boxed{T_{r,N}(a,\eta) = \eta \det_{0\leq i,j<q}\left(\sum_{n=0}^{N} n^{i+j} L_a(n)\{(1+\eta)(n+a)+1\}\right).} \tag{4.3}$$

The determinant of size zero is one.

Proof. The harmonic pencil is $C = uu^{\mathsf{T}} + \eta vv^{\mathsf{T}}$, where $u_i = (-a)^i$ and $v_i = (-a-1)^i$. Its common kernel consists of coefficient vectors of polynomials divisible by L_a. Use the basis

$$1,\ n,\ L_a(n),\ nL_a(n), \dots, n^{q-1}L_a(n).$$

Its monic triangular change of basis has determinant one. The upper-left two-by-two minor of C has determinant η, while the lower-right compression is a sum of polynomial weights:

$$R_\eta(n;a)L_a(n)^2 = L_a(n)\{(1+\eta)(n+a)+1\}.$$

Selecting the two harmonic entries in the transformed determinant proves (4.3). □

Define the monic uniform Hahn polynomial by the finite expression

$$p_k(x;N) = \binom{2k}{k}^{-1} \sum_{j=0}^{k} (-1)^{k+j} \binom{k}{j} \binom{k+j}{j} (N-j)^{\underline{k-j}} x^{\underline{j}}. \tag{4.4}$$

For $0 \le k \le N$ it is orthogonal on $\{0, \dots, N\}$ with unit weights. Its squared norm is

$$h_k(N) = \sum_{n=0}^{N} p_k(n;N)^2 = \frac{(N+k+1)!}{(2k+1)(N-k)!\binom{2k}{k}^2}, \qquad \mathcal{H}_q(N) = \prod_{k=0}^{q-1} h_k(N). \tag{4.5}$$

The finite formula also avoids apparent singularities in the usual terminating hypergeometric notation

$$p_k(x;N) = \frac{(-1)^k N^{\underline{k}}}{\binom{2k}{k}} {}_3F_2\left(\begin{matrix} -k, k+1, -x \\ 1, -N \end{matrix}; 1\right).$$

Theorem 4.2 (A fixed-size Hahn formula)**.** *Let $r \ge 2$, $N \ge r$, $\eta \ne -1$, and put $t = 1+\eta$ and*

$$c_1 = -a, \qquad c_2 = -a-1, \qquad c_3 = -a - \frac{1}{t}, \qquad q = r-2.$$

Then

$$\boxed{T_{r,N}(a,\eta) = (-1)^q t^r \mathcal{H}_q(N) \det_{1\le i\le 3,\ 0\le j\le 2} p_{q+j}(c_i;N).} \tag{4.6}$$

The expression has a removable limit at $\eta = -1$ and a zero at $\eta = 0$. The compressed formula (4.3) specifies those cases without a limit. Thus a determinant of arbitrary size has a highest harmonic coefficient represented by one determinant of size three.

Proof. For monic orthogonal polynomials p_k of a measure ν, let $D_q[\nu]$ be its moment determinant. For distinct $c_1, \dots, c_d$, the Christoffel transformation is

$$D_q\left[\prod_{\ell=1}^{d}(x-c_\ell)\nu\right] = (-1)^{dq} D_q[\nu] \frac{\det(p_{q+j-1}(c_i))_{i,j=1}^{d}}{\prod_{i<j}(c_j - c_i)}. \tag{4.7}$$

One way to verify the sign and normalization is to append the d fixed nodes c_i to the q integration nodes in the Vandermonde determinant, replace its monomial columns by the monic orthogonal columns, and integrate the first q rows against their companion Vandermonde. Orthogonality leaves the product of the first q squared norms; the remaining d rows give the displayed determinant. The cross factors are $\prod_{\ell,j}(c_\ell - x_j)$, producing $(-1)^{dq}$. This also proves (4.7) for a finite discrete measure by finite Cauchy–Binet.

In (4.3), the weight is $t\prod_{\ell=1}^{3}(n-c_\ell)$. Moreover

$$\prod_{i<j}(c_j - c_i) = \frac{\eta}{t^2}.$$

Substitution in (4.7), followed by multiplication by the exterior η, gives $(-1)^{3q} t^{q+2} \mathcal{H}_q \det p$, which is (4.6). The proof initially assumes $\eta \ne 0, -1$; the extension follows from the polynomial identity (4.3). □

4.2 Every exceptional residue parameter is an orthogonal zero

The complex degree theorem allows a finite exceptional set of cutoffs for a fixed weight. Here one can solve the dual question: for a fixed cutoff, which pole residues cause degree loss?

Corollary 4.3 (Exact exceptional parameters)**.** *Let $a > 0$, $q = r - 2 \geq 1$, and $N \geq q$. Let π_q be the monic orthogonal polynomial of degree q for the positive discrete measure with weights $L_a(n)$ on $\{0, \dots, N\}$, and let $\xi_1, \dots, \xi_q$ be its zeros. These zeros are distinct and belong to $(0, N)$. Put*

$$\mathcal{H}_q[L_a] = \det_{0 \leq i,j < q} \left(\sum_{n=0}^{N} n^{i+j} L_a(n) \right) > 0.$$

Then

$$T_{r,N}(a, \eta) = \eta \, \mathcal{H}_q[L_a] \prod_{j=1}^{q} \{(1+\eta)(a+\xi_j) + 1\}. \tag{4.8}$$

Apart from $\eta = 0$, where the pencil rank changes, all degree-deficient parameters are exactly

$$\eta_j = -1 - \frac{1}{a + \xi_j} \quad (1 \leq j \leq q). \tag{4.9}$$

They are simple, real, and less than -1. In particular no nonreal residue parameter causes degree loss at these cutoffs.

Proof. The one-factor case of (4.7), with $c = -a - 1/(1+\eta)$, gives $D_q[((1+\eta)(n+a)+1)L_a] = \mathcal{H}_q[L_a] \prod_j \{(1+\eta)(a+\xi_j)+1\}$. The standard zero theorem for a positive orthogonal measure gives simplicity and the location [8, Section 18.2(vi)]. Equation (4.3) finishes the proof. □

These statements concern formal harmonic degree. They do not assert that the evaluated determinant has a zero, and they do not specify how far its degree drops beyond the vanishing of the quadratic term. The nonreal example in Section 3 shows why this distinction matters.

4.3 A uniform classical-polylogarithm formula

Now specialize to $a = 1$ and use $M = N + 1$. Passing from monomials in n to monomials in $k = n + 1$ preserves the determinant. Write $S_j(M) = \sum_{k=1}^{M} k^j$, regarded as its Faulhaber polynomial, and let X be the formal replacement for H_M. The moment of degree $j \geq 0$ is

$$\mu_j(M, X; \eta) = \delta_{j0} X + \mathbf{1}_{j \geq 1} S_{j-1}(M) + \eta \left\{ \sum_{\ell=0}^{j-1} (-1)^{j-1-\ell} S_\ell(M) + (-1)^j \left(X + \frac{1}{M+1} - 1 \right) \right\}. \tag{4.10}$$

The sum is empty at $j = 0$. This is simply polynomial division of $k^j/(k+1)$. Consequently

$$\mathcal{P}_r(M, X; \eta) = \det(\mu_{i+j})_{0 \leq i,j < r}$$

is an explicit finite construction of every cutoff formula.

Theorem 4.4 (Only a dilogarithm is needed for every r)**.** *For $r \geq 2$ there are polynomials $A_{rj}(M; \eta)$, $0 \leq j \leq 2$, such that*

$$\mathcal{P}_r(M, X; \eta) = \sum_{j=0}^{2} A_{rj}(M; \eta) X^j + \frac{B_r(X; \eta)}{M+1}, \tag{4.11}$$

where the entire endpoint residue is

$$B_2 = \eta(X - \eta - 2), \qquad B_3 = -\eta, \qquad B_r = 0 \ (r \geq 4). \tag{4.12}$$

Let $\theta = x\,d/dx$, $L = -\log(1-x)$, *and*

$$F_0(x) = \frac{x}{1-x}, \qquad F_1(x) = \frac{L}{1-x}, \qquad F_2(x) = \frac{\mathrm{Li}_2(x) + L^2}{1-x}.$$

For $|x| < 1$ *the cutoff generating function is*

$$\boxed{\sum_{M\geq 1} Z_{r,M-1}(1,\eta)x^M = \sum_{j=0}^{2} A_{rj}(\theta;\eta)F_j(x) + \mathcal{R}_r(x;\eta),} \tag{4.13}$$

with

$$\begin{aligned} \mathcal{R}_2 &= \eta\left\{\frac{L^2}{2x} - (\eta+2)\left(\frac{L}{x} - 1\right)\right\}, \\ \mathcal{R}_3 &= -\eta\left(\frac{L}{x} - 1\right), \qquad \mathcal{R}_r = 0 \quad (r \geq 4). \end{aligned} \tag{4.14}$$

All apparent singularities at $x = 0$ *are removable. For every particle count the result therefore belongs to the space over* $\mathbb{C}(x)$ *generated by* $1, L, L^2, \mathrm{Li}_2(x)$.

Proof. In the k basis put $e_0 = (1,0,\ldots)^{\mathsf{T}}$, $e_1 = (0,1,0,\ldots)^{\mathsf{T}}$, and $v_i = (-1)^i$. The matrix in (4.10) has the form

$$A(M) + X(e_0 e_0^{\mathsf{T}} + \eta v v^{\mathsf{T}}) + \frac{\eta v v^{\mathsf{T}}}{M+1},$$

where A is polynomial in M. The harmonic part has rank at most two, and the pole has rank one. The determinant thus has degree at most two in X and at most a simple pole at $M = -1$.

The polynomial identities $S_0(-1) = -1$ and $S_j(-1) = 0$ for $j \geq 1$ show that the regular matrix at that point is

$$X(e_0 e_0^{\mathsf{T}} + \eta v v^{\mathsf{T}}) - \eta e_0 e_0^{\mathsf{T}} - (e_0 e_1^{\mathsf{T}} + e_1 e_0^{\mathsf{T}}).$$

Its residue after the rank-one update is directly $\eta(X - \eta - 2)$ at $r = 2$ and $-\eta$ at $r = 3$. For $r \geq 4$, the matrix even after adding an arbitrary multiple of vv^{T} has image contained in $\mathrm{span}(e_0, e_1, v)$, of dimension at most three. Its determinant is identically zero, so the pole residue vanishes. Polynomial division proves (4.11).

The elementary harmonic generating identities are

$$\sum_{M\geq 1} H_M x^M = F_1(x), \qquad \sum_{M\geq 1} H_M^2 x^M = F_2(x).$$

For the second, split H_M^2 into its diagonal terms and twice its strict off-diagonal terms, then sum the outer cutoff. Multiplication by a polynomial in M becomes its substitution at θ. Finally

$$\sum_{M\geq 1} \frac{x^M}{M+1} = \frac{L}{x} - 1, \qquad \sum_{M\geq 1} \frac{H_M x^M}{M+1} = \frac{L^2}{2x};$$

the latter follows by integrating F_1. These facts prove the formula and convergence. Differentiation by θ preserves the displayed four-function space, since $d\,\mathrm{Li}_2(x)/dx = L/x$. □

This is a closed finite recipe, with polynomial coefficient construction (4.10). It does not claim that the four functions are always a minimal basis. Special parameters can produce further cancellations. Nor does formal harmonic degree in a general problem automatically equal minimal multiple-polylogarithm depth; the conclusion here uses the specific pole cancellation proved above.

4.4 A positive three-particle harmonic identity

The choice $\eta = -1$ makes the weight positive and especially simple:

$$R_{-1}(n;1) = \frac{1}{(n+1)(n+2)}.$$

Let

$$\mathcal{Z}_N = \sum_{0\le n_1<n_2<n_3\le N} \frac{\prod_{i<j}(n_j-n_i)^2}{\prod_{i=1}^{3}(n_i+1)(n_i+2)}.$$

The first two values vanish. Direct evaluation of the three-by-three moment determinant in (4.10) gives the following identity for every integer $M \ge 1$:

$$\boxed{\mathcal{Z}_{M-1} = -\frac{M(M+1)(M+2)}{3}H_M^2 + \frac{4M^2(M+2)}{3}H_M + \frac{M^3(M^2-18M-31)}{12(M+1)}.} \tag{4.15}$$

For example $\mathcal{Z}_2 = 1/36$ and $\mathcal{Z}_3 = 23/90$. Equation (4.15) is also an exact digamma formula after $H_M = \psi(M+1)+\gamma$.

Its entire cutoff generator has a short classical expression. With $B = \mathrm{Li}_2(x) + L^2$,

$$\boxed{\begin{aligned} G(x) := \sum_{M\ge1} \mathcal{Z}_{M-1}x^M = &-\frac{2xB}{(1-x)^4} + \frac{(3-14x+15x^2-2x^3)L}{3x(1-x)^4} \\ &+ \frac{18x^3-31x^2+31x-6}{6(1-x)^5}. \end{aligned}} \tag{4.16}$$

The right side starts $x^3/36 + 23x^4/90 + \cdots$ despite its apparent low-order terms. One proof is to apply Theorem 4.4 to (4.15); it reduces the statement to differentiating F_0, F_1, F_2 a finite number of times. Thus both the sum and the displayed closed form are established without a numerical relation search.

4.5 A convergent zeta-valued endpoint

The determinant is $O(M^4 + M^3\log^2 M)$, so its sixth Euler primitive has an ordinary finite value at one.

Corollary 4.5 (An explicit ensemble integral)**.**

$$\begin{aligned} \mathcal{E} &:= \sum_{M\ge1} \frac{\mathcal{Z}_{M-1}}{M^6} = \frac{1}{120}\int_0^1 \frac{(-\log x)^5}{x} G(x)\,dx \\ &= -1 + \frac{13}{12}\zeta(2) - \frac{31}{12}\zeta(3) + \frac{5}{3}\zeta(4) + \frac{41}{6}\zeta(5) - \frac{97}{24}\zeta(6) - 4\zeta(7) \\ &\quad - \frac{7}{3}\zeta(2)\zeta(3) + 2\zeta(3)^2 + \frac{2}{3}\zeta(2)\zeta(5) + \frac{5}{3}\zeta(3)\zeta(4). \end{aligned} \tag{4.17}$$

Numerically $\mathcal{E} = 0.00273146713162708026426923909 78\ldots$.

Proof. The elementary integral $\int_0^1 x^{M-1}(-\log x)^5\,dx = 5!/M^6$ and positivity justify interchange by Tonelli. Set $T_s = \sum_{M\ge1} H_M/M^s$ and $S_s = \sum_{M\ge1} H_M^2/M^s$. Equation (4.15) and the partial fraction

$$\frac{M^2-18M-31}{12M^3(M+1)} = \frac{1}{M+1} - \frac{1}{M} + \frac{13}{12M^2} - \frac{31}{12M^3}$$

give

$$\mathcal{E} = -\frac{S_3 + 3S_4 + 2S_5}{3} + \frac{4}{3}(T_3 + 2T_4) - 1 + \frac{13}{12}\zeta(2) - \frac{31}{12}\zeta(3). \tag{4.18}$$

The classical Euler-sum reductions required here are

$$\begin{aligned} T_3 &= \tfrac{5}{4}\zeta(4), & T_4 &= 3\zeta(5) - \zeta(2)\zeta(3), \\ S_3 &= \tfrac{7}{2}\zeta(5) - \zeta(2)\zeta(3), & S_4 &= \tfrac{97}{24}\zeta(6) - 2\zeta(3)^2, \\ S_5 &= 6\zeta(7) - \zeta(2)\zeta(5) - \tfrac{5}{2}\zeta(3)\zeta(4). \end{aligned} \tag{4.19}$$

Substitution proves the result. For completeness, the next paragraph derives these classical inputs, so the endpoint evaluation does not depend on a table of numerical relations. □

Write $\zeta(s_1, \ldots, s_d) = \sum_{k_1 > \cdots > k_d \geq 1} k_1^{-s_1} \cdots k_d^{-s_d}$. Directly sorting the harmonic indices gives

$$\begin{aligned} T_s &= \zeta(s,1) + \zeta(s+1), \\ S_s &= 2\zeta(s,1,1) + \zeta(s,2) + 2\zeta(s+1,1) + \zeta(s+2). \end{aligned} \tag{4.20}$$

These are particular cases of the established Euler-sum/MZV calculus [11, 12]. The height-one generating identity, also classical, is

$$\begin{aligned} \sum_{n\geq 2,\ m\geq 0} \zeta(n, \{1\}^m) X^{n-1} Y^{m+1} &= 1 - \frac{\Gamma(1-X)\Gamma(1-Y)}{\Gamma(1-X-Y)} \\ &= 1 - \exp \sum_{k\geq 2} \frac{\zeta(k)}{k} \{X^k + Y^k - (X+Y)^k\}. \end{aligned} \tag{4.21}$$

A self-contained proof starts from $\sum_m H_{N-1}(\{1\}^m) Y^m = \prod_{j<N}(1 + Y/j)$. Sum first over n and then N, to obtain

$$Y \int_0^1 (t^{-X} - 1)(1-t)^{-1-Y}\, dt = Y\{B(1-X, -Y) - B(1, -Y)\}.$$

Initially take $\Re Y < 0$ and X, Y small; the beta integral gives the gamma quotient and analytic continuation gives the identity at zero. Its coefficients yield Euler's formula

$$\zeta(n,1) = \frac{n}{2}\zeta(n+1) - \frac{1}{2}\sum_{j=1}^{n-2} \zeta(j+1)\zeta(n-j)$$

and, at the orders needed in (4.20),

$$\begin{aligned} \zeta(3,1,1) &= 2\zeta(5) - \zeta(2)\zeta(3), \\ \zeta(4,1,1) &= \tfrac{23}{16}\zeta(6) - \zeta(3)^2, \\ \zeta(5,1,1) &= 5\zeta(7) - 2\zeta(2)\zeta(5) - \tfrac{5}{4}\zeta(3)\zeta(4). \end{aligned}$$

The remaining double zeta values follow from the convergent stuffle identity $\zeta(a)\zeta(b) = \zeta(a,b) + \zeta(b,a) + \zeta(a+b)$ and the iterated-integral shuffle identity

$$\zeta(a)\zeta(b) = \sum_{j=0}^{a-1} \binom{b-1+j}{j} \zeta(b+j, a-j) + \sum_{j=0}^{b-1} \binom{a-1+j}{j} \zeta(a+j, b-j).$$

Use $(a,b)=(2,3),(3,3),(2,5),(3,4)$ and the height-one values just obtained. The resulting finite linear systems give

$$\begin{aligned}\zeta(3,2)&=3\zeta(2)\zeta(3)-\tfrac{11}{2}\zeta(5),\\ \zeta(4,2)&=\zeta(3)^2-\tfrac{4}{3}\zeta(6),\\ \zeta(5,2)&=5\zeta(2)\zeta(5)+2\zeta(3)\zeta(4)-11\zeta(7).\end{aligned}$$

Together with the elementary even-zeta products these prove (4.19). We claim the ensemble specialization and its resulting integral as the additional identities here; the Euler-sum values and the gamma generating identity themselves are established background.

5 Unequal-rate Stieltjes boundary collisions

The independent-shift product generator in the incoming *Lattice Contacts and Mixed Zeta Identities* treats positive shifts and proves compatibility with the coincident finite part along the common-shift diagonal. Its Question Q9, repeated at the end of its independent-shift section [6, Section 6 and Question Q9], leaves open the path

$$a=\lambda\epsilon,\qquad b=\mu\epsilon,\qquad \lambda,\mu>0,\qquad \epsilon\downarrow 0.$$

We determine every divergent power and logarithm on this path and its remaining constant, for arbitrary Stieltjes indices and argument derivatives. The constant correction is considerably smaller than the full independent-shift product generator: it depends only on two Taylor coefficients at one and a finite polynomial in the shift ratio.

These statements resolve the specified collision question relative to the inspected repository sources. The underlying local-subtraction method is classical; no claim of historical priority for general finite-part asymptotics is intended.

5.1 The coordinate and the finite contact polynomial

We use the convention

$$Z(u,z)=\zeta(1+u,z)-\frac{1}{u}=\sum_{m\ge 0}\frac{(-1)^m}{m!}\gamma_m(z)u^m. \tag{5.1}$$

In particular, $\gamma_0(z)=-\psi(z)$. Write

$$I^{r,t}_{m,n}(\epsilon;\lambda,\mu)=\int_0^1\gamma_m^{(r)}(x+\lambda\epsilon)\gamma_n^{(t)}(x+\mu\epsilon)\,dx,\qquad m,n,r,t\in\mathbb{Z}_{\ge 0}. \tag{5.2}$$

Every integral in (5.2) is ordinary. Its asymptotic constant means the coefficient of $\epsilon^0(\log\epsilon)^0$. The inherited coincident finite part is

$$F^{r,t}_{m,n}=\mathrm{CT}_{\delta\downarrow 0}\int_\delta^1\gamma_m^{(r)}(x)\gamma_n^{(t)}(x)\,dx. \tag{5.3}$$

Thus the two constants use the displayed coordinates ϵ and δ; no subtraction scale is left implicit.

For $r\ge 0$ define a holomorphic germ at $u=0$ by

$$C_r(u;\lambda,\mu)=\frac{(-1)^r(1+u)_r}{r!}\left\{\frac{\lambda^{-u}-1}{u}+\lambda^{-u}\sum_{j=1}^{r}\binom{r}{j}\frac{(\mu/\lambda-1)^j}{u+j}\right\}, \tag{5.4}$$

and put

$$\kappa_{m,r}(\lambda,\mu) = (-1)^m\, \partial_u^m C_r(u;\lambda,\mu)|_{u=0}\,. \tag{5.5}$$

The sum is empty if $r = 0$. The quotient at $u = 0$ is removable. Moreover, $(1+u)_r/(u+j)$ is a polynomial for $1 \le j \le r$, so (5.4) contains no hidden special-function limit.

Theorem 5.1 (Complete constant correction for unequal rates)**.** *For all indices in* (5.2)*,*

$$\boxed{\begin{aligned}\mathrm{CT}_{\epsilon\downarrow 0}\, I_{m,n}^{r,t}(\epsilon;\lambda,\mu) = F_{m,n}^{r,t} &+ \gamma_n^{(r+t)}(1)\kappa_{m,r}(\lambda,\mu)\\ &+ \gamma_m^{(r+t)}(1)\kappa_{n,t}(\mu,\lambda).\end{aligned}} \tag{5.6}$$

In particular, with $R = \mu/\lambda$*,*

$$\boxed{\kappa_{0,r}(\lambda,\mu) = (-1)^r \left\{ -\log\lambda + \sum_{j=1}^{r} \frac{R^j - 1}{j} \right\}.} \tag{5.7}$$

For arbitrary Stieltjes indices without argument differentiation,

$$\mathrm{CT}_{\epsilon\downarrow 0}\, I_{m,n}^{0,0} = F_{m,n}^{0,0} - \frac{\gamma_n(1)}{m+1}(\log\lambda)^{m+1} - \frac{\gamma_m(1)}{n+1}(\log\mu)^{n+1}. \tag{5.8}$$

Only the Taylor coefficient of degree r in the analytic part of the second factor and the coefficient of degree t in the analytic part of the first factor enter (5.6). In particular, the singular–singular product changes the divergent terms but contributes no path-dependent constant.

5.2 A finite formula for every contact coefficient

For a fully finite harmonic description, write

$$A_r(u) = \frac{(1+u)_r}{r!} = \sum_{a=0}^{r} e_{r,a} u^a, \qquad Q_r(u;R) = \sum_{j=1}^{r} \binom{r}{j} (R-1)^j \frac{A_r(u)}{u+j} = \sum_{a=0}^{r-1} q_{r,a}(R) u^a.$$

The Q_0 sum is empty. The coefficients $e_{r,a}$ are elementary symmetric polynomials in $1, 1/2, \ldots, 1/r$, equivalently

$$A_r(u) = \exp\left\{ \sum_{h\ge 1} \frac{(-1)^{h+1}}{h} H_r^{(h)} u^h \right\}. \tag{5.9}$$

With $L = \log\lambda$, coefficient extraction in (5.4) gives

$$\begin{aligned}\kappa_{m,r} = (-1)^{m+r} m! &\sum_{a=0}^{\min(r,m)} e_{r,a} \frac{(-L)^{m-a+1}}{(m-a+1)!}\\ &+ (-1)^{m+r} m! \sum_{a=0}^{\min(r-1,m)} q_{r,a}(R) \frac{(-L)^{m-a}}{(m-a)!}.\end{aligned} \tag{5.10}$$

The second sum is interpreted as empty at $r = 0$. For example,

$$\kappa_{0,0} = -L, \qquad \kappa_{1,0} = -\frac{L^2}{2},$$

$$\kappa_{0,1} = L + 1 - R, \qquad \kappa_{1,1} = \frac{L^2}{2} - RL,$$

$$\kappa_{0,2} = -L + R + \frac{R^2}{2} - \frac{3}{2}.$$

At $\lambda = 1$ the contact is a polynomial of degree at most $r-1$ in u. Consequently $\kappa_{m,r}(1,\mu) = 0$ whenever $m \ge r$. At $\lambda = \mu = 1$ all contacts vanish. These are exact truncations of the correction, not asymptotic statements about its size.

5.3 The complete singular power–logarithm expansion

The following evaluated germ specifies every divergent term as well as the constant. Put

$$P_r(u) = (-1)^r(1+u)_r, \qquad Z_r(u,x) = \partial_x^r Z(u,x) = P_r(u)x^{-r-1-u} + A_r^{\mathrm{reg}}(u,x), \tag{5.11}$$

where $A_r^{\mathrm{reg}}(u,x) = \partial_x^r Z(u,1+x)$. The superscript distinguishes this analytic function from the finite polynomial A_r above. Define

$$a_{t,k}(v) = \frac{1}{k!}\partial_z^{t+k} Z(v,z)\big|_{z=1}. \tag{5.12}$$

These are holomorphic near $v=0$, and

$$(-1)^n \partial_v^n a_{t,k}(0) = \frac{\gamma_n^{(t+k)}(1)}{k!}.$$

The only nonpolynomial local kernel needed is the elementary beta integral

$$\begin{aligned} B_{r,t}(u,v;\lambda,\mu) &= P_r(u)P_t(v)\int_0^\infty \frac{dy}{(y+\lambda)^{r+1+u}(y+\mu)^{t+1+v}} \\ &= \frac{P_r(u)P_t(v)\mu^{-r-t-1-u-v}}{r+t+1+u+v} \\ &\quad \cdot {}_2F_1\left(\begin{matrix} r+1+u,\ r+t+1+u+v \\ r+t+2+u+v \end{matrix}; 1-\frac{\lambda}{\mu}\right). \end{aligned} \tag{5.13}$$

The integral fixes its branch. It is holomorphic on a neighborhood of $(u,v)=(0,0)$, including $\lambda=\mu$.

For $0 \le k < r$, define the finite expression

$$M_{r,k}(u;\lambda,\mu) = P_r(u)\lambda^{k-r-u}\sum_{j=0}^{k}\binom{k}{j}\frac{(\mu/\lambda-1)^j}{r-k+j+u}. \tag{5.14}$$

For the critical degree use the removable germ

$$N_r(u;\epsilon,\lambda,\mu) = P_r(u)\left\{\epsilon^{-u}\lambda^{-u}\left(\frac{1}{u}+\sum_{j=1}^{r}\binom{r}{j}\frac{(\mu/\lambda-1)^j}{u+j}\right) - \frac{1}{u}\right\}. \tag{5.15}$$

At each fixed $\epsilon>0$ its apparent pole at $u=0$ cancels before any differentiation. Its constant in $\log\epsilon$ is $r!C_r(u;\lambda,\mu)$.

Finally let $\mathcal{F}_{r,t}(u,v)$ denote the holomorphic germ whose coefficient dictionary is

$$(-1)^{m+n}\partial_u^m\partial_v^n\mathcal{F}_{r,t}(0,0) = F_{m,n}^{r,t}. \tag{5.16}$$

To specify it independently of a meromorphic regularization, set

$$\begin{aligned} R_{r,t}(u,v;x) &= Z_r(u,x)Z_t(v,x) - P_r(u)P_t(v)x^{-r-t-2-u-v} \\ &\quad - P_r(u)\sum_{k=0}^{r}a_{t,k}(v)x^{k-r-1-u} - P_t(v)\sum_{k=0}^{t}a_{r,k}(u)x^{k-t-1-v}, \end{aligned} \tag{5.17}$$

and define

$$\begin{aligned} \mathcal{F}_{r,t}(u,v) = &\int_0^1 R_{r,t}(u,v;x)\,dx - \frac{P_r(u)P_t(v)}{r+t+1+u+v} \\ &- P_r(u)\sum_{k=0}^{r-1}\frac{a_{t,k}(v)}{r-k+u} - P_t(v)\sum_{k=0}^{t-1}\frac{a_{r,k}(u)}{t-k+v}. \end{aligned} \tag{5.18}$$

Empty sums are zero. The critical terms $k=r$ and $k=t$ have no added constant: every monomial $x^{-1}\log^h x$ has finite part zero on $(0,1)$ in the coordinate (5.3). This is why retaining a whole meromorphic $1/u$ expression would give the wrong Stieltjes coefficient convention.

Theorem 5.2 (All singular terms in one joint germ)**.** *For fixed $r, t \geq 0$ and $\lambda, \mu > 0$, as $\epsilon \downarrow 0$,*

$$\begin{aligned}\int_0^1 Z_r(u, x+\lambda\epsilon) Z_t(v, x+\mu\epsilon)\, dx &= \mathcal{F}_{r,t}(u,v) + \epsilon^{-r-t-1-u-v} B_{r,t}(u, v; \lambda, \mu) \\ &+ \sum_{k=0}^{r-1} a_{t,k}(v) \epsilon^{k-r-u} M_{r,k}(u; \lambda, \mu) + \sum_{k=0}^{t-1} a_{r,k}(u) \epsilon^{k-t-v} M_{t,k}(v; \mu, \lambda) \\ &+ a_{t,r}(v) N_r(u; \epsilon, \lambda, \mu) + a_{r,t}(u) N_t(v; \epsilon, \mu, \lambda) + o(1). \end{aligned} \tag{5.19}$$

The remainder tends to zero locally uniformly for (u, v) in a complex neighborhood of zero. Every fixed mixed order derivative may therefore be taken in this asymptotic formula. Applying $(-1)^{m+n}\partial_u^m \partial_v^n$ and setting $u = v = 0$ gives the complete negative-power and logarithmic expansion of (5.2), through its constant.

The possible negative powers are especially sparse. They consist of $\epsilon^{-(r+t+1)}$, together with $\epsilon^{-r}, \ldots, \epsilon^{-1}$ and $\epsilon^{-t}, \ldots, \epsilon^{-1}$ when the corresponding derivative orders are positive. The first power has a logarithmic polynomial of degree at most $m + n$. The two finite mixed families have degrees at most m and n. The critical terms give logarithmic polynomials of degrees at most $m + 1$ and $n + 1$. Formula (5.19) includes cancellations between these polynomials automatically.

Proof. The Hurwitz recurrence gives (5.11) exactly. Its analytic part has the Taylor expansion

$$A_t^{\mathrm{reg}}(v, z) = \sum_{k=0}^{r} a_{t,k}(v) z^k + O(z^{r+1})$$

uniformly for v in a sufficiently small closed complex disk. Expand the product of its singular and analytic parts. For the singular–singular term, substitute $x = \epsilon y$ and complete the interval to $(0, \infty)$. Its omitted tail tends locally uniformly to $P_r(u)P_t(v)/(r + t + 1 + u + v)$, giving precisely the beta term in (5.19) and the first rational constant in (5.18).

For a mixed Taylor monomial the finite identity is

$$\begin{aligned} &P_r(u) \int_0^1 (x+\lambda\epsilon)^{-r-1-u} (x+\mu\epsilon)^k\, dx \\ &\quad = P_r(u) \sum_{j=0}^{k} \binom{k}{j} ((\mu-\lambda)\epsilon)^j \frac{(\lambda\epsilon)^{k-j-r-u} - (1+\lambda\epsilon)^{k-j-r-u}}{r-k+j+u}. \end{aligned} \tag{5.20}$$

It follows by expanding $x + \mu\epsilon = (x + \lambda\epsilon) + (\mu - \lambda)\epsilon$ and integrating each power. At $k < r$, the lower endpoint is $\epsilon^{k-r-u} M_{r,k}$, while the upper endpoint tends to $-P_r(u)/(r - k + u)$. At $k = r$ the combination of the lower endpoint and $-P_r(u)/u$ is exactly N_r. The remaining upper-endpoint error is holomorphic at $u = 0$ and tends uniformly to zero; in particular, $\{1 - (1 + \lambda\epsilon)^{-u}\}/u = O(\epsilon)$ uniformly near zero. Exchange the two factors for the other mixed terms.

It remains to justify passage to the limit in the Taylor remainders. For $|\Re u|, |\Re v| \leq \eta < 1/4$ and fixed positive rates, a singular factor times the Taylor remainder is bounded by $C(x+\epsilon)^{-\eta}$, hence by $Cx^{-\eta}$ on $(0, 1)$. This integrable majorant is uniform on compact parameter disks. The product of analytic remainders is uniformly bounded. Dominated convergence therefore gives the integral in (5.18), locally uniformly in u, v. Cauchy's formula on slightly larger parameter disks gives the same conclusion after every prescribed mixed order derivative. This proves (5.19), including its joint rather than merely raywise spectral meaning.

Coefficient extraction in (5.18) gives (5.16): noncritical singular monomials have their ordinary finite primitives at one, whereas every critical $x^{-1} \log^h x$ contributes zero. Thus no exchange between an unsubtracted divergent integral and a spectral derivative is used. □

Proof of Theorem 5.1. In (5.19), the beta term and the noncritical mixed terms have strictly negative powers of ϵ, also after spectral differentiation. They cannot contribute to the constant under discussion. In each critical term, retain the constant in $\log \epsilon$, namely $r!C_r$ or $t!C_t$. Equation (5.12) then gives (5.6). At $u = 0$,

$$C_r(0; \lambda, \mu) = (-1)^r \left\{ -\log \lambda + \sum_{j=1}^{r} \binom{r}{j} \frac{(R-1)^j}{j} \right\}.$$

The polynomial identity

$$\sum_{j=1}^{r} \binom{r}{j} \frac{(R-1)^j}{j} = \sum_{j=1}^{r} \frac{R^j - 1}{j}$$

follows by differentiating both sides in R and using their common value zero at $R = 1$. This proves (5.7). At $r = 0$, expanding $(\lambda^{-u} - 1)/u$ gives (5.8). □

5.4 Polylogarithmic coefficients of the divergent beta term

The hypergeometric notation in (5.13) does not add an unspecified family of constants. It is enough to differentiate the base kernel in the shifts:

$$B_{r,t}(u, v; \lambda, \mu) = \partial_\lambda^r \partial_\mu^t B_{0,0}(u, v; \lambda, \mu).$$

For $\lambda > \mu$, put $d = \lambda - \mu$ and $z = d/\lambda$. The substitution $y = d/(x + \lambda)$ gives

$$B_{0,0}(u, v; \lambda, \mu) = d^{-1-u-v} \int_0^z \frac{\exp\{(u+v)\log y - v \log(1-y)\}}{1-y}\, dy. \tag{5.21}$$

Every coefficient of total spectral degree N is a finite linear combination of harmonic polylogarithms on the alphabet $\{0, 1\}$ and logarithms, of weight at most $N + 1$. This follows by shuffling the logarithmic factors and integrating once with $dy/(1 - y)$. Shift differentiation preserves this finite function class. The case $\mu > \lambda$ follows by exchange; the coincident case is the elementary expression

$$B_{r,t}(u, v; \lambda, \lambda) = \frac{P_r(u) P_t(v) \lambda^{-r-t-1-u-v}}{r + t + 1 + u + v}.$$

Thus every coefficient in the full subtraction belongs to an explicit finite alphabet of harmonic numbers, logarithms, polylogarithms, and Hurwitz or Stieltjes derivatives at one.

5.5 Three completely explicit polygamma collision laws

Write

$$D(\lambda, \mu) = \frac{\log(\lambda/\mu)}{\lambda - \mu}, \qquad D(\lambda, \lambda) = \frac{1}{\lambda}, \tag{5.22}$$

and let D_λ and $D_{\lambda\mu}$ be its indicated derivatives. Explicitly, away from the diagonal,

$$D_\lambda = \frac{(\lambda - \mu)/\lambda - \log(\lambda/\mu)}{(\lambda - \mu)^2},$$
$$D_{\lambda\mu} = \frac{(\lambda - \mu)(\lambda^{-1} + \mu^{-1}) - 2\log(\lambda/\mu)}{(\lambda - \mu)^3}.$$

Their diagonal values are $-1/(2\lambda^2)$ and $1/(3\lambda^3)$. The inherited coincident evaluations [6] are

$$F_{0,0}^{0,0} = 2\gamma_1 - 2\zeta(2), \qquad F_{0,0}^{1,0} = \zeta(2), \qquad F_{0,0}^{1,1} = 2\zeta(3) + 4\zeta'(3).$$

Using these established values in the new collision formula gives

$$\int_0^1 \psi(x+\lambda\epsilon)\psi(x+\mu\epsilon)\,dx = \frac{D(\lambda,\mu)}{\epsilon} - 2\gamma\log\epsilon + 2\gamma_1 - 2\zeta(2) - \gamma\log(\lambda\mu) + o(1), \tag{5.23}$$

$$\int_0^1 \psi'(x+\lambda\epsilon)\psi(x+\mu\epsilon)\,dx = \frac{D_\lambda(\lambda,\mu)}{\epsilon^2} - \frac{\gamma}{\lambda\epsilon} + \boxed{\zeta(2)\left\{\frac{\mu}{\lambda} + \log\frac{\mu}{\lambda}\right\}} + o(1), \tag{5.24}$$

$$\begin{aligned}\int_0^1 \psi'(x+\lambda\epsilon)\psi'(x+\mu\epsilon)\,dx = {} & \frac{D_{\lambda\mu}(\lambda,\mu)}{\epsilon^3} + \frac{\zeta(2)}{\epsilon}\left(\frac{1}{\lambda}+\frac{1}{\mu}\right) + 4\zeta(3)\log\epsilon \\ & + 4\zeta'(3) + 2\zeta(3)\left\{3 + \log(\lambda\mu) - \frac{\mu}{\lambda} - \frac{\lambda}{\mu}\right\} + o(1).\end{aligned} \tag{5.25}$$

The ordered law (5.24) is invariant under a common scaling of the two rates. Its exchange companion has constant $\zeta(2)\{\lambda/\mu + \log(\lambda/\mu)\}$. Their sum equals $\zeta(2)(\lambda/\mu + \mu/\lambda)$, in exact agreement with the endpoint derivative of the digamma product. This independently checks the logarithmic signs and the ratio terms; exchange symmetry alone would not determine either ordered constant separately.

5.6 A first Stieltjes collision with a dilogarithm

Let $\gamma_j = \gamma_j(1)$ and put

$$E(\lambda,\mu) = \frac{\log(\lambda/\mu)\log\lambda + \mathrm{Li}_2(1-\mu/\lambda)}{\lambda-\mu}, \qquad E(\lambda,\lambda) = \frac{1+\log\lambda}{\lambda}. \tag{5.26}$$

This is the ordinary convergent integral of $\log(y+\lambda)/\{(y+\lambda)(y+\mu)\}$ on $(0,\infty)$, as follows either from (5.21) or one integration by parts. The standard coincident product germ gives

$$F_{1,0}^{0,0} = \frac{3}{2}\gamma_2 + 2\gamma\zeta(2) - \zeta(3). \tag{5.27}$$

Consequently the complete first-index expansion is

$$\begin{aligned}\int_0^1 \gamma_1(x+\lambda\epsilon)\gamma_0(x+\mu\epsilon)\,dx = {} & \frac{D(\lambda,\mu)\log\epsilon + E(\lambda,\mu)}{\epsilon} - \frac{\gamma}{2}(\log\epsilon)^2 \\ & - (\gamma\log\lambda + \gamma_1)\log\epsilon + \frac{3}{2}\gamma_2 + 2\gamma\zeta(2) - \zeta(3) \\ & - \frac{\gamma}{2}(\log\lambda)^2 - \gamma_1\log\mu + o(1).\end{aligned} \tag{5.28}$$

All logarithms and the dilogarithm in this formula are real for positive rates, with the displayed continuous diagonal limit.

For completeness, (5.27) can be extracted without a numerical relation search. Put $w = u+v$, $Q = \Gamma(1+w)/\{\Gamma(1+u)\Gamma(1+v)\}$ and $E_0(u) = \pi u/\sin\pi u$. The inherited coincident Hurwitz product formula gives the holomorphic finite-part germ

$$\begin{aligned}\mathcal{F}_{0,0}(u,v) = {} & \frac{w - Q\{uE_0(v) + vE_0(u)\}}{uvw} \\ & + \frac{Z(v,1) - QE_0(u)Z(w,1)}{u} + \frac{Z(u,1) - QE_0(v)Z(w,1)}{v}.\end{aligned}$$

Each divided difference is removable. Its degree-zero term is $2\gamma_1 - 2\zeta(2)$ and its coefficient of u is $\zeta(3) - 3\gamma_2/2 - 2\gamma\zeta(2)$. Applying the sign convention (5.1) proves (5.27).

5.7 Verification, scope, and further questions

The accompanying verifier checks the finite mixed primitive, the rising-factorial cancellations, the compact ratio polynomial, and contact jets by two exact coefficient constructions. Separate high-precision ordinary quadratures check the three beta kernels and the three polygamma collisions at diagonal, unequal integral, and unequal rational rates. The numerical collision residuals are compared at decreasing positive ϵ after the complete displayed subtraction. They test the asymptotic constants and normalization; they are not certified error enclosures.

The result concerns fixed positive rates. Paths with a vanishing or unbounded ratio require an additional asymptotic regime; inserting a rate that depends on ϵ into a fixed-compactness proof would not justify that regime. The following are concrete next questions.

1. Determine the complete transition for $a = \lambda\epsilon^{\alpha}$, $b = \mu\epsilon^{\beta}$ with $\alpha \neq \beta$, including the boundary between the two scales. The explicit finite ratio polynomial and the beta kernel make every potential extra power visible.

2. Extend the locality theorem to three independently shifted Stieltjes factors. With more than two factors, products of two singular parts and one analytic Taylor coefficient can already affect the constant; the required kernel should be stated before any reduction to a smaller alphabet is claimed.

3. Classify linear combinations of ordered derivative products whose collision constant depends only on the ratio μ/λ, or is independent of the rates. Formula (5.10) reduces this to an exact finite polynomial problem at each index.

4. Continue the rates to a common complex sector and derive the jumps when the lower-endpoint logarithms cross their chosen cuts. The local beta integral supplies a natural branch convention.

6 Source corrections, integration decisions, and further research

6.1 Precise changes to the source status

No newly false inherited mathematical theorem was found in the proof dependencies used here. The proposed changes are extensions of valid theorems and corrections to research-status lists. They should be applied with their original attribution intact.

1. In the ordered-boundary report's Section 9, replace the statement that complete gap-six coverage is open by a reference to Theorem 2.1, explicitly recording its computation-assisted proof. Retain the broader all-pairs question. The source's stronger sign conjecture $f_b > 0$ for every $b \geq 7$ is not proved by a nonzero finite-field residue. We establish positivity for $b \geq 2048$ and nonvanishing at all earlier admissible indices; we make no additional finite sign claim.

2. In the bilinear determinant report's Section 10, mark the complex rational-weight degree–rank question solved by Theorem 3.3. Broaden the generic rational-weight degree statement to complex coefficients. Preserve the real eventual-sign hypotheses in the source's sharper numerical exception bound; that bound does not follow from the new proof.

3. In the lattice-contact report's Question Q9 and the final paragraph of its independent-shift section, mark the fixed positive unequal-rate collision question solved by Theorems 5.1 and 5.2. Keep nonlinear paths and ratios tending to zero or infinity open.

4. Do not reopen the already proved smaller-order-ten and smaller-order-twelve cases. Do not relabel the arbitrary mixed even-tail product calculus or the existing Stieltjes spectral-face results as new consequences of this article.

Several implementation conventions deserve explicit retention. In the late Bernoulli coefficient matrix, a lower binomial index of zero is omitted, rather than evaluated as $\binom{n}{0} = 1$. Changing that convention changes the theorem's matrix. In the complex rank proof the relevant bilinear operation is ordinary transposition; replacing it by conjugate transposition would change the rational algebraic problem. In the Stieltjes section the constant depends on the stated small parameter; critical Laurent quotients must be removed jointly before taking spectral derivatives. These are guards against incorrect integrations of the results, not allegations that the cited source theorems made those mistakes.

The manuscript's old rejected S_8 candidate should remain rejected, and its revised S_8 and short S_6 candidates should retain their conjectural status. Nothing in the present rank, Hahn, or collision arguments implies those period equalities. A small numerical residual does not supply their missing proof.

6.2 A proposed research program

The next questions below are selected for their potential to produce exact identities or structural theorems. Each has a concrete starting point supplied by the article.

Q1. Complete the gap-eight family. For half-gap $m = 4$, make the inherited four-mode Gamma basis effective. Two modes have a vanishing leading imaginary phase. Derive a uniform two-term estimate for each, control the four-by-four limiting determinant, and obtain an explicit finite completion threshold. The target is nonvanishing for every $b \geq 5$, not simply a larger numerical table.

Q2. Replace the gap-six finite completion by a structural proof. Find a recurrence invariant or a variation-diminishing theorem which proves $f_b \neq 0$ at every admissible b. A fixed-sign assertion from the first index cannot work because the initial sign transition is real. The stronger source conjecture $f_b > 0$ for $b \geq 7$ would also settle the sign issue left unresolved by the modular certificate.

Q3. Connect fixed-gap and fixed-smaller-order regimes. The source has eventual theorems in both directions, and the present proof makes one gap ray effective. Develop a two-parameter determinant expansion uniform when both indices increase. The limiting reciprocal-square nodes become less separated as the gap grows; merely reusing fixed-gap constants does not cover the intervening region.

Q4. Evaluate the Gamma normalization determinants. The initial determinant is nonzero by an integral alternant argument. Seek explicit Gamma, hypergeometric, or recurrence evaluations for its normalization K_m at $m = 2, 3, \ldots$. Such evaluations would turn the recurrence certificates into more compact identities rather than just nonvanishing tests.

Q5. Classify rationally summable polynomial subspaces. For a given rational weight R, classify the spaces V on which every discrete-residue bilinear form of Rfg vanishes. Theorem 3.2 proves nondegeneracy of every rational primitive once V is known; it does not classify these spaces. This asks for a concrete bridge between rational summability and symmetric pencils with varying kernels.

Q6. Find intrinsic complex cutoff-exception invariants. Express the degree or factorization of the exceptional-cutoff polynomial directly from the pole orders, residues, and

integer dispersions of R, without constructing the full determinant. The coefficient-gcd criterion is exact, but a residue-level description could explain and predict the later nonreal example in Section 3.

Q7. Extend the adjacent-pole compiler to several poles. For $R(n) = \sum_{\ell=1}^{d} c_\ell/(n + a_\ell)$, determine when the highest harmonic coefficient reduces to a determinant of fixed size as r varies. For integer-separated poles the compressed measure is a polynomial transform, suggesting a Hahn or multiple-orthogonal representation. The difficult part is to classify endpoint pole cancellations and exceptional residue varieties, rather than invoke the general Christoffel formula alone.

Q8. Classify when classical polylogarithms suffice. Theorem 4.4 gives an entire arbitrary-rank family using only one dilogarithm. Determine necessary and sufficient conditions on a rational weight for its cutoff generators to remain in a fixed classical-polylogarithm space. Generic harmonic degree supplies an upper-level clue but is not by itself a minimal-depth theorem.

Q9. Evaluate the Euler primitives of the ensemble family. For the explicit $\mathcal{Z}_N$, determine all endpoint values $\sum_{M\geq 1} \mathcal{Z}_{M-1}/M^s$ in their convergence range, and their alternating companions. Equation (4.18) generalizes to a small list of linear and quadratic Euler sums. The question is when these reduce to ordinary zeta products and when irreducible higher weight components first survive.

Q10. Resolve genuinely different collision scales. Derive the full expansion for $a = \lambda\epsilon^\alpha$ and $b = \mu\epsilon^\beta$, $\alpha \neq \beta$. The fixed-ratio theorem cannot simply be used with an ϵ-dependent ratio without new uniform analysis. Its beta kernel and finite contact polynomial identify the terms which must enter a transition formula.

Q11. Develop three-factor collision constants. For three independently shifted Stieltjes derivatives, products of two singular factors with one analytic coefficient may affect the constant. Determine the finite local contact tensor and the appropriate multivariable beta kernel before claiming a reduction to the two-factor alphabet.

Q12. Identify invariant ordered combinations and complex jumps. Classify linear combinations of derivative products whose collision constants depend only on the rate ratio, or are independent of both rates. Then continue the rates in a common complex sector and compute logarithmic jumps across the boundary cuts. At fixed orders the first task is an explicit finite polynomial problem in the contact coefficients.

The strongest unresolved global targets remain the unrestricted moment-prefix conjecture and the manuscript's distinguished Gaussian period identities. The present article supplies three completed local research programs and concrete next steps toward broader statements; it does not infer those global targets from finite evidence.

A The Gamma basis used in the gap-six proof

This appendix reproduces the case $m = 3$ of the Gamma-basis theorem in the incoming *Ordered Boundary Identities and Gamma Kernels* report [3, Section 8]. Its purpose is to make the analytic input to Theorem 2.1 independently reproducible within this article. The contour construction and the initial determinant argument are inherited results; the effective estimates and complete gap-six conclusion are established in Section 2.

Put

$$W(x) = \left|\Gamma\left(\tfrac{1}{4} + \tfrac{ix}{2}\right)\right|^{12}, \quad P_1(x) = \sinh(2\pi x), \quad P_2(x) = \sinh(\pi x), \quad P_3(x) = \tanh(\pi x).$$

For $1 \le j \le 3$, define

$$G_j(z) = 4\sin(z/2)\int_0^\infty \sinh(xz)P_j(x)W(x)\,dx, \qquad t_{n,j} = [z^{2n}]G_j(z). \tag{A.1}$$

The density estimate of Lemma 2.2, or the ordinary Gamma modulus asymptotic [8, Equation 5.11.9], proves locally uniform convergence in $|\operatorname{Re} z| < j\pi$. These functions are even and vanish at zero.

Proposition A.1 (Specialized Gamma basis)**.** *The three germs in* (A.1) *satisfy*

$$(G_j^{(6)} - G_j^{(6)}(0))\cot(z/2) + (G_j\cot(z/2))^{(6)} = 0. \tag{A.2}$$

Their sixth derivatives at zero are

$$G_1^{(6)}(0) = G_2^{(6)}(0) = 0, \qquad G_3^{(6)}(0) = 256\pi^3. \tag{A.3}$$

The matrix $T = (t_{n,j})_{1\le n,j\le 3}$ has negative determinant. Thus these germs form a basis of the formal null space of (A.2)*, and their later coefficient rows satisfy*

$$(t_{b+i,j})_{i,j=1}^3 = F_b T$$

with the rational fundamental matrix of Section 2.

A.1 The Fourier shift and its two residues

For one of the three odd multipliers P, set

$$\mathcal{F}_P(t) = \int_{\mathbb{R}} e^{ixt}P(x)W(x)\,dx, \qquad \mathcal{Q}_P(t) = 2\sinh(t/2)\mathcal{F}_P(t), \qquad D = \frac{d}{dt}.$$

The integrals and all derivatives converge absolutely for real t and define analytic germs. The oddness of P makes $\mathcal{F}_P$ odd, so $\mathcal{Q}_P$ is even and vanishes at zero. Define

$$\mathcal{E}_P(t) = e^t(D+\tfrac12)^6\mathcal{F}_P(t) - e^{-t}(D-\tfrac12)^6\mathcal{F}_P(t).$$

Since

$$\mathcal{Q}_P = (e^{t/2} - e^{-t/2})\mathcal{F}_P, \qquad \mathcal{Q}_P\coth(t/2) = (e^{t/2} + e^{-t/2})\mathcal{F}_P,$$

ordinary differentiation gives the exact identity

$$\coth(t/2)\mathcal{Q}_P^{(6)}(t) + (\mathcal{Q}_P(t)\coth(t/2))^{(6)} = \frac{\mathcal{E}_P(t)}{\sinh(t/2)}. \tag{A.4}$$

For complex arguments, use the single-valued meromorphic continuations

$$W(y) = \left[\Gamma(\tfrac14 + iy/2)\Gamma(\tfrac14 - iy/2)\right]^6,$$
$$V(y) = \left[\Gamma(\tfrac34 + iy/2)\Gamma(\tfrac34 - iy/2)\right]^6.$$

The absolute-value notation for W is used only on the real axis. Gamma recurrence gives

$$(iy - \tfrac12)^6 W(y+i) = (iy + \tfrac12)^6 W(y-i) = 64V(y). \tag{A.5}$$

In the first Fourier integral defining $\mathcal{E}_P$, substitute $y = x - i$, so its contour is $\mathbb{R} - i$; in the second substitute $y = x + i$, so its contour is $\mathbb{R} + i$. These substitutions remove the respective

$e^{\pm t}$ factors. Moreover $P(y+i) = P(y-i)$ for every allowed P. Thus bottom minus top is the integral of the same meromorphic function

$$64e^{iyt}V(y)P(y+i) \tag{A.6}$$

on the two horizontal sides of a positively oriented rectangle.

The nearest poles of V are $\pm 3i/2$, outside the closed strip $|\operatorname{Im} y| \le 1$. The apparent closer Gamma poles in the shifted W expressions have already been canceled exactly by (A.5). For a sinh multiplier, $P(y+i)$ is an entire function: it equals $\sinh(2\pi y)$ for P_1 and $-\sinh(\pi y)$ for P_2. For P_3 it equals $\tanh(\pi y)$, with precisely two poles in the strip, $y = \pm i/2$. Uniform Stirling estimates on the closed strip bound the integrand by a polynomial times $e^{-(3-k)\pi|\operatorname{Re} y|}$ in the sinh case, where $k = 2$ or 1, and by a polynomial times $e^{-3\pi|\operatorname{Re} y|}$ in the tanh case. For fixed real t, the factor e^{iyt} is bounded there. The vertical sides therefore vanish as their real parts tend to infinity. All contour shifts are consequently justified.

There are no enclosed poles for P_1, P_2, so $\mathcal{E}_{P_1} = \mathcal{E}_{P_2} = 0$. At each tanh pole the residue is $1/\pi$, and

$$V(i/2) = V(-i/2) = \pi^3.$$

The residues of (A.6) are therefore $64\pi^2 e^{-t/2}$ and $64\pi^2 e^{t/2}$. Bottom minus top is $2\pi i$ times their sum, giving

$$\mathcal{E}_{P_3}(t) = 256i\pi^3 \cosh(t/2). \tag{A.7}$$

The second term on the left of (A.4) is holomorphic at zero, since $\mathcal{Q}_P(t) = O(t^2)$. The first has Laurent coefficient $2\mathcal{Q}_P^{(6)}(0)$ at t^{-1}. Comparing (A.4) with (A.7) yields

$$\mathcal{Q}_{P_1}^{(6)}(0) = \mathcal{Q}_{P_2}^{(6)}(0) = 0, \qquad \mathcal{Q}_{P_3}^{(6)}(0) = 256i\pi^3.$$

The exact identities also show that

$$(\mathcal{Q}_P^{(6)} - \mathcal{Q}_P^{(6)}(0))\coth(t/2) + (\mathcal{Q}_P \coth(t/2))^{(6)} = 0.$$

Pairing positive and negative x gives $G_j(z) = i\mathcal{Q}_{P_j}(iz)$. Rotation uses $\coth(iz/2) = -i\cot(z/2)$, proving (A.2). Since $i \cdot i^6 = -i$, it also gives (A.3).

A.2 The initial determinant has an exact positive integral

Write $\phi_n(x) = \operatorname{Im}(x + i/2)^{2n}$. The elementary identity

$$\sin(z/2)\sinh(xz) = \sum_{n\ge 1} \frac{\phi_n(x)}{(2n)!} z^{2n}$$

and local uniform domination imply

$$t_{n,j} = \frac{4}{(2n)!}\int_0^\infty \phi_n(x)P_j(x)W(x)\,dx.$$

The polynomial ϕ_n has leading term nx^{2n-1} and only lower odd powers. A triangular row transformation consequently gives

$$\det T = \left(\prod_{n=1}^{3} \frac{4n}{(2n)!}\right) \det\left(\int_0^\infty x^{2n-1}P_j(x)W(x)\,dx\right)_{n,j=1}^{3}, \qquad \prod_{n=1}^{3}\frac{4n}{(2n)!} = \frac{1}{90}.$$

For $\Delta(y_1, y_2, y_3) = \prod_{i<j}(y_j - y_i)$, continuous Cauchy–Binet gives the following exact normalization:

$$\boxed{\begin{aligned}\det T = &-\frac{1}{270}\int_{(0,\infty)^3} \Delta(x_1^2, x_2^2, x_3^2) \\ &\cdot \Delta(\cosh \pi x_1, \cosh \pi x_2, \cosh \pi x_3) \prod_{r=1}^{3} x_r W(x_r) \tanh(\pi x_r)\, d\boldsymbol{x} < 0.\end{aligned}} \tag{A.8}$$

Indeed $\det(x_r^{2n-1}) = (\prod_r x_r)\Delta(x_1^2, x_2^2, x_3^2)$. After dividing row r of $\det(P_j(x_r))$ by $\tanh(\pi x_r)$ and putting $v_r = \cosh(\pi x_r)$, its columns are $2v_r^2, v_r, 1$. Its determinant is $-2\Delta(v_1, v_2, v_3)$; together with the Cauchy–Binet factor $1/3!$ and $1/90$ this gives $-1/270$. The integral converges absolutely by exponential decay. Both Vandermondes have the same sign whenever the x_r are distinct, since $x \mapsto x^2$ and $x \mapsto \cosh(\pi x)$ are strictly increasing on $(0, \infty)$. Its integrand is therefore positive off the collision planes, proving strict negativity.

The coefficient recurrence of Section 2 gives a three-dimensional formal null space with arbitrary first three coefficients. Equation (A.8) proves that our three germs span it, and coefficient comparison proves $(t_{b+i,j}) = F_b T$, completing Proposition A.1. As a normalization check, (A.3) also gives the inherited ordinary Gamma moments

$$\int_0^\infty \left(3x^5 - \frac{5}{2}x^3 + \frac{3}{16}x\right) P_j(x) W(x)\, dx = \begin{cases} 0, & j = 1, 2, \\ 64\pi^3, & j = 3. \end{cases}$$

B An exact $(14, 20)$ polygamma-tail evaluation

For $(p, q) = (14, 20)$ put $K(y) = y^{21} \sum_{j=0}^{9} c_j y^{2j}$, where the coefficient row $(c_0, \ldots, c_9)$ is

$$\begin{aligned} -44227285070991635910 0, &\quad -916343865300090386578 \\ 7827122405129750301156, &\quad -17591879878937090990204 \\ 2401562621096548985816 0, &\quad -27581453408920271748009 \\ 5150817385609631175057 2, &\quad 5326403712175170016988 2 \\ 9020686868591908422037 2, &\quad 346718746623385922565 \end{aligned}$$

Then the centered difference $P(x) = K(x + 1/2) - K(x - 1/2)$ and the polynomial

$$\begin{aligned} S(x) = &\frac{1040156239870157767695}{19}x^6 + \frac{1618580484745959817903323}{494}x^4 \\ &+ \frac{171589487161427755057692583}{3952}x^2 + \frac{259647741562565023962318 52}{247}. \end{aligned}$$

give the exact absolutely convergent identity

$$\begin{aligned} \sum_{n\geq 0} \left\{ \frac{P(n+1/2)}{13!\,19!}\psi^{(13)}(n+1)\psi^{(19)}(n+1) - S(n+1/2) \right\} \\ = -\frac{70897571597555316074547}{2}. \end{aligned}$$

The complete finite telescoper is recorded in `gap_six_14_20_example.json` and regenerated by `create_gap_six_example.py`. Its verification uses only polynomial and rational-function identities followed by the displayed tail asymptotics.

C Reproducibility, proof dependencies, and verification

C.1 What is supplied

The package contains the modular article source, an assembled single-file TEX source, the compiled PDF, the exact finite-field certificate, an explicit $(14, 20)$ polynomial identity with its finite telescoper, and portable verification programs. A claim ledger separates proved results, inherited inputs, numerical diagnostics, and remaining conjectures. The source manifest pins every archived dependency to the repository revision and a SHA-256 hash.

The main analytic dependency for gap six is the three-dimensional Gamma basis and its complementary-minor normalization in the ordered-boundary contribution [3]. The recurrence and normalization are restated in the main proof; Appendix A proves the specialized Gamma contour and initial-determinant assertions. The unchanged inherited source is included for inspection under `provenance/inherited/ordered_gamma_basis.tex`. The earlier complementary-minor source [7] and the two source sections posing the complex-weight and independent-shift questions are also preserved. They are provenance artifacts, and are not silently incorporated as new authorship.

The new algebraic primitive theorem and its degree consequence are proved directly in Section 3. The Hahn transformation and Euler reductions are derived at the needed normalization in Section 4. The entire unequal-rate asymptotic is obtained by finite local subtraction in Section 5; its coincident constants are attributed to the inherited product formula and independently checked.

C.2 Exact computations and their roles

Program	What it establishes or checks
`verify_gap_six.py`	The finite part of the proof: 2,044 nonzero determinants modulo $2^{31} - 1$, with trial-division primality and invertible denominators. Also 2,049 independent cotangent coefficients and 17 exact original-matrix normalization checks.
`create_gap_six_example.py`	Generates the integer coefficient row, sextic subtraction, and complete finite telescoper for $(p, q) = (14, 20)$. Checks exact rational differences and the limiting constant.
`verify_complex_rank.py`	66 exact assertions: rank-one minors, complex isotropic updates, rational harmonic decompositions, exceptional cutoff identities, and a varying-kernel pencil.
`verify_hahn_polylogs.py`	149 exact assertions: monic Hahn orthogonality and norms, highest coefficients, compressed determinants, ensemble sums, pole residues, and generator coefficients. Separate 75-digit quadratures test the generator and zeta endpoint.
`derive_euler_endpoint.py`	21 exact coefficient assertions derived from the height-one gamma quotient and convergent shuffle/stuffle relations. No numerical relation search is used.
`verify_collisions.py`	119 exact assertions about primitives and contact coefficients. Separate 65-digit beta and polygamma quadratures check the complete collision subtraction at several rates.

Program	What it establishes or checks
Two independent audit scripts	Reconstruct original Bernoulli matrices, check the cofactor normalization and exact threshold constants, and derive the sensitive Stieltjes and trigamma finite parts independently.

The modular certificate is finite and exhaustive on the stated interval, because the analytic proof has already reduced the unresolved indices to that interval. It would not prove an unbounded conjecture without that analytic step. Conversely, the high-precision integrations are not used to assert that an unproved residual vanishes.

The independent audits rederived the complex rank-one minor, all high-risk constants in the gap-six argument, the leading inverse-row cofactor sign, and the locally uniform Stieltjes remainder. In particular they recovered $F_{1,0}^{0,0} = 3\gamma_2/2 + 2\gamma\zeta(2) - \zeta(3)$ and $F_{0,0}^{1,1} = 2\zeta(3) + 4\zeta'(3)$ separately from the collision derivation. Audit notes and their exact output are included. This is independent internal checking, not external peer review or a formal proof-assistant certificate.

C.3 Rebuild instructions

The build is local and requires no network calls. In the package directory run

```
python -m pip install -r requirements.txt
python verification/run_all.py
python build.py
```

The verification dependencies are Python, SymPy, and mpmath. The PDF build uses a standard `pdflatex` installation with the packages listed in `article.tex`. The build program assembles the standalone TEX file and runs three passes for the table of contents and references. It does not execute shell escape.

The verification runner records each exit status and elapsed time and stops on a failure. The result JSON files record their scope and arithmetic mode. Numerical values can vary in their last diagnostic digits with library versions; the exact finite-field residue list is identified by its canonical hash. The package manifest checks artifact integrity, while the recurrences and the supplied programs make the mathematical certificate independently reproducible.

References

[1] ProveIt project, *Polylogarithm manuscript*. Repository revision `c4036d60715916ae72dd18983782fdb182840b86`. Pinned manuscript directory. Inspected 11–12 October 2026.

[2] ProveIt project, *Incoming research contributions*, same revision. Pinned incoming directory. The full eleven-archive inventory and hashes are in the accompanying provenance record.

[3] ProveIt research continuation, *Ordered Boundary Identities and Gamma Kernels*, 11 October 2026. Especially Section 8, *The Gamma basis and eventual fixed-gap reconstruction*, and Section 9, research status and questions. Pinned archive. The Gamma-basis source used in this article is included unchanged under `provenance/inherited/`.

[4] ProveIt research continuation, *Bilinear Alternating Tails and Harmonic Determinants*, 11 October 2026. Especially `sections/05_degree_rank.tex` and `sections/10_research_questions.tex`. Pinned archive.

[5] ProveIt research continuation, *Bilinear Tail Identities, Polylogarithmic Resonances, and Stieltjes Boundary Products*, 11 October 2026. Pinned archive. This contribution already includes the smaller-order-twelve result.

[6] ProveIt research continuation, *Lattice Contacts and Mixed Zeta Identities*, 11 October 2026. Especially Sections 4 and 6, and Question Q9 in `sections/08_audit_and_research.tex`. Pinned archive.

[7] ProveIt research continuation, *Harmonic Tail Gamma*: `sections/02_gap_four.tex`. Pinned recurrence and complementary-minor source.

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